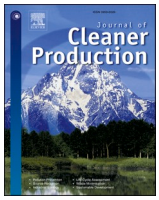
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Review



Multi-scale solar radiation and photovoltaic power forecasting with machine learning algorithms in urban environment: A state-of-the-art review

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ABSTRACT

Solar energy has been rapidly utilized in urban environments owing to its significant potential to fulfill the energy demand. The precise forecasting of solar energy, including solar radiation and photovoltaic power forecasting, is crucial for effective energy utilization in cities. Currently, artificial intelligence algorithms, including machine learning (ML) and deep learning (DL) algorithms, have been widely utilized in the research of solar energy forecasting and have shown excellent performance. The different research scales lead to variations in the influencing factors and forecasting approaches used in solar energy prediction. Therefore, this paper provides a comprehensive review of solar radiation and photovoltaic power forecasting research using ML and DL algorithms from a multi-scale perspective, including the meso-scale, micro-scale, and building-scale forecasting in urban environments. The aim is to summarize the state-of-the-art progress and evaluate the solar forecasting effectiveness in diverse research scenarios. The characteristic analyses of ML- and DL-based solar forecasting approaches, prediction models, forecasting horizons, inputs and outputs were conducted for different research scales. This paper discusses the existing issues and future research directions for multiscale solar energy forecasting in cities. The analyses can help researchers and engineers in identifying suitable prediction algorithms and approaches. Consequently, this exploration can contribute to the efficient utilization of solar radiation and photovoltaic power generation in multi-scale urban environments.

1. Introduction

Solar energy, as a primary renewable energy source, is utilized because of its significant potential to fulfill the energy demand and the limited availability of fossil fuels worldwide (Ağbulut et al., 2021). Solar energy, available on the earth's surface, can receive approximately 1.5 to 10^{18} KWh/year of solar energy annually (Gupta et al., 2021; Sun et al., 2018). Consequently, the photovoltaic (PV) market has steadily increased to produce solar power over the past years because of the supported national policy, renewable national energy targets, and reduced cost of PV modules (Mellit and Kalogirou, 2021). As a result, PV utilization has witnessed a significant global expansion in scale and electricity generation, with the capacity increasing from 6.6 GW to over 500 GW during 2006–2018 (Gernaat et al., 2020). In recent years, there has been a growing focus on the development of PV power stations as well as building integrated with photovoltaics (BIPV). Hence, the accurate solar energy forecasting is becoming increasingly necessary before the design and installation of solar systems.

The precise forecasting of solar radiation and PV power is highly desirable to increase its availability in the urban environment (Anderson and Leach, 2004). Although solar energy is considered a promising resource, it poses several threats when it integrates into the power grids (Bella Espinar, José-Luis Aznarte et al., 2010). The main challenge of solar energy is its variability and dependence on weather conditions, which makes it different from conventional power sources in the sense that future production can be precisely planned. The prediction of energy consumption (Li et al., 2023; Wang et al., 2021), energy demand (Wang et al., 2018a), oil production and consumption (Q. Wang et al., 2022d; Wang and Song, 2019), carbon emission reduction (Q. Wang et al., 2022e), and shale gas production (Wang et al., 2018b) has become increasingly important in recent years. Hence, it is essential to precisely predict the demand and supply of power (Kumari and Toshniwal, 2021).

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Relevant studies showed that a 25% improvement in PV power output forecast accuracy could lead to a reduction of 1.56% (US$ 46.5 million) in the net generation cost (Yagli et al., 2019). Therefore, solar radiation and PV power forecasting are vital for evaluating the solar energy potential of a region.

Solar radiation is composed of essential components, including the Direct Normal Irradiation (DNI), Diffuse Horizontal Irradiation (DHI), and Global Horizontal Irradiation (GHI). The GHI is the summation of DNI and DHI (Gupta et al., 2021; Kumari and Toshniwal, 2021). Solar radiation has significant implications for various applications, including the PV power generation, PV plant installation, demand and supply balancing, and storage management (Frías-Paredes et al., 2017). Solar radiation is measured using instruments, such as meteorological stations. However, these instruments are only available at some stations worldwide because of the high installation and maintenance costs, as well as calibration requirements. For instance, the available meteorological stations that can provide the measured GHI data are 122 and 43 in China and India, respectively (Zang et al., 2012). Furthermore, solar radiation is highly influenced by the location and climate. Hence, it is important to predict solar radiation values with meteorological and geographical parameters, such as the sunshine duration, temperature, relative humidity, cloud cover, latitude, and longitude. Solar radiation plays a crucial role in PV power (H. Wang et al., 2020b). Therefore, numerous methodologies have been proposed for solar radiation and PV power prediction using time-series datasets, diverse forecasting horizons and models.

In addition, the solar utilization potential of blocks and buildings has been researched owing to the increasing use of PV modules in urban environments. Notably, homeowners with rooftop PV systems are the primary drivers of the significant increase in PV capacity, accounting for the majority of the installed capacity today (The IEA PVPS Technology Collaboration Program, 2023). Homeowners can follow the policy instrument to sell the excess electricity to the grid at the retail price, which can benefit them from subsidies (Gernaat et al., 2020). Hence, there is great significance in the solar potential forecasting of buildings, which includes the evaluation of solar radiation, PV power, and economic potential. For forecasting the solar potential of a city, the PV installation potential of large-scale rooftops has been widely researched. By utilizing aerial images and imagery segmentation models, the rooftop outlines can be extracted to calculate the solar potential. This approach provides policymakers with valuable information for large-scale solar utilization planning on building roofs. It can offer overall PV suggestions regarding the annual rooftop PV power potential of a region. Furthermore, the solar potential forecasting of neighborhoods and individual buildings has been studied to obtain more accurate predictions for both roofs and façades. Unlike large-scale forecasting, these two smaller scales can consider the interactions between buildings, such as shading and distance factors. The simulation software enables the determination of solar radiation distribution on building roofs and facades. Depending on different urban conditions, PV installation strategies for building surfaces can be proposed. However, owing to the complexity of the urban environment and the challenges in modeling, a few studies have trained numerous block or building models and predicted solar potential using machine learning approaches.

Thus far, many studies have been conducted to suggest a precise solar energy prediction in cities using two main approaches: Artificial Intelligence (AI) and conventional approaches (Lima et al., 2020). Conventional approaches include using physical modeling, statistical methods, and regression approaches to predict energy (Sharma and Kakkar, 2018). These approaches are most effective for linear problems. On the other hand, AI methods work well with nonlinear outputs, which have become very popular among almost all engineering fields in recent decades (Lima et al., 2020). AI is an emerging technology that utilizes computer simulations to study human logical thinking, reasoning, and group behavior (Ciulla et al., 2019). Among its various algorithms, machine learning (ML) stands out as a rapidly developing branch. The idea of ML is to learn patterns from large datasets and use these patterns to predict unknown data. Through learning from mistakes and comparing errors, ML can efficiently explore the possibilities of datasets, allowing for significant advancements in self-domain exploitation (Zhou, 2022).

Machine learning usually deals with classification and regression problems, with prevalent algorithms including Random Forest, Support Vector Machine, and K-Nearest Neighbors. Among ML algorithms, deep learning (DL) employs a deep neural network structure to learn hierarchical features from datasets, resulting in more efficient representations (Bedi and Toshniwal, 2019; Lima et al., 2020; Moncada et al., 2018). DL techniques can deal with data-driven, time-sequence, and image-based datasets, utilizing Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN). AI algorithms with ML/DL approaches provides several advantages (Khosravi et al., 2019): (1) they possess powerful feature extraction and nonlinear mapping functions that can reveal deeper patterns in datasets; (2) they exhibit high compatibility and can be flexibly embedded into various solar energy forecasting scenarios, such as solar radiation and PV power forecasting; (3) they can achieve a certain degree of logical reasoning and realize the intelligence of solar predictors. Therefore, ML/DL algorithms with excellent forecasting performance have found widespread applications in solar energy prediction.

1.1. Contributions of this work

Existing solar prediction primarily focuses on forecasting solar radiation and PV power, which have been conducted on multiple scales. Some research predicts the spatial-temporal solar radiation and PV power on a large scale, and some research predicts the distribution and utilization of solar radiation and PV power potential on building surfaces on a small scale. The different scales of research lead to variations in the influencing factors and forecasting approaches used in solar energy prediction. Therefore, this paper aims to provide a comprehensive review of state-of-the-art multiscale solar radiation and photovoltaic power forecasting in the urban environment using ML and DL algorithms. It evaluates the effectiveness and applicability of predictions in diverse research scenarios. The novelties and contributions of this study are as follows.

(1) Existing ML- and DL-based solar radiation and photovoltaic power forecasting approaches have been extensively reviewed from a multi-scale perspective, including the meso-scale, micro-scale, and building-scale forecasting in the urban environment.
(2) An analysis of ML- and DL-based solar radiation and photovoltaic power forecasting approaches, prediction models, input and output variables, and forecasting horizons in the datasets are conducted. The analysis results can provide researchers with suitable forecasting approaches and models for multiscale solar utilization and forecasting scenarios in cities.
(3) The characteristics and key points of solar radiation and photovoltaic power forecasting at multiple scales, as well as future research potential, are explored. The findings serve to inspire researchers in addressing unresolved issues in solar utilization and development in cities.

In summary, this paper presents a review of ML- and DL-based solar radiation and photovoltaic power forecasting in the urban environment, examining analysis approaches, prediction models, and dataset characteristics from a multi-scale viewpoint. Section 2 discusses the meso-scale, micro-scale, and building-scale solar energy forecasting. Section 3 analyzes the ML and DL approaches, including the input and target data, data resources, machine and deep learning models, and forecasting horizons. Sections 4 to 6 delve into detailed discussions on meso-scale, micro-scale, and building-scale solar forecasting research. Section 7 presents an outlook on future directions, while the main conclusions are

summarized in Section 8.

2. Methodology

The analysis proposed in this study was developed using a multi-scale methodology for forecasting solar radiation and photovoltaic power, based on a spatial-temporal model. Initially, the query string was employed in Google Scholar and Elsevier Scopus databases to obtain a wider range of relevant studies. The query string combined terms related to solar forecasting (prediction, solar irradiation, photovoltaic power, solar potential) and artificial intelligence (machine learning, deep learning) and retrieved relevant results.

For research on building energy, different scales have distinct concerns and research approaches. Research on the solar energy prediction analyzes multiple time-series data at different scales. Therefore, it is necessary to establish clear multi-scale classification criteria and categorize all papers included in this study. Fig. 1 shows a summary of multi-scale solar radiation and photovoltaic power forecasting with machine learning models between 2001 and 2022. A total of 115 studies were identified in this review. The use of ML algorithms for solar energy forecasting has been increasing since 2019. Overall, the research mainly focuses on the meso-scale solar radiation and power forecasting, which accounts for 63.5% of the total studies, followed by the building-scale (20%) and micro-scale (16.5%).

There are multiple spatial scales in the research of urban physics and energy analysis, which provide an overview of the spatial-temporal scales of atmospheric phenomena (Blocken, 2015; Dogan and Reinhart, 2017). In the area of solar energy prediction research, the analysis of the literature included in this study mainly focuses on three scales: meso-scale, micro-scale, and building-scale. Table 1 and Table 2 show information on multi-scale solar energy forecasting research, including the scale, size, and database. The classification of the research scale is based on the differences in the target value, sample size, database, etc. The main prediction target with the general sample size for each scale is shown. The column displays the horizontal size and distance of the research scales. The databases show the primary data resources at different scales.

In addition, the prediction and analysis of the three research scales have certain similarities and differences, forming the basis for the multi-scale classification of all literature in this study. Each research scale has its independent research approach and similar internal logic, as shown in Fig. 2. In general, the overall research approach of each scale is similar, whereas different scales make diverse selections in each part of the prediction process. The target data for multiscale research mainly comprise solar radiation and solar power prediction, leading to various input data selections for each scale. These inputs are derived from different resources, such as the collected data, the simulated data from software, and the calculated data from a polynomial. Once the dataset is prepared, all the input and target data are imported into the prediction models, which contain deep learning, machine learning, and hybrid models. For the prediction, the literature chooses a different prediction horizon. Most studies adopt diverse prediction models and explore multiple combinations and optimization algorithms to achieve the optimal prediction performance.

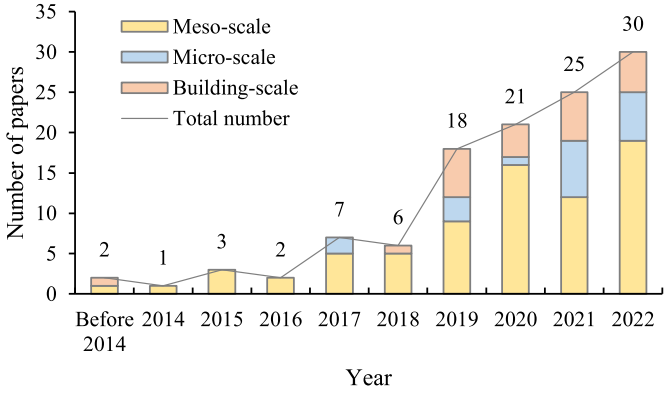


Fig. 1. Summary of research trends in multi-scale solar radiation and photovoltaic power forecasting with machine learning models.

2.1. Meso-scale solar energy forecasting

For the meso-scale solar energy forecasting research, the horizontal size is generally less than 200 km, and the sample size is always more than 2000 with experiential data (Table 1). Therefore, this research primarily involves three fields: overall solar radiation forecasting, solar power forecasting, and large-scale rooftop solar photovoltaic potential forecasting using spatial-temporal datasets (Table 2, Table 3).

The research on solar radiation forecasting investigates predictions on meteorological parameters and radiation data in various climate regions, to determine the parameters that most significantly impact the distribution of solar radiation. Solar radiation is affected by weather factors, seasonal factors, pollution factors, etc. In addition, solar radiation directly impacts the solar power performance of PV panels. Hence, the research on solar power forecasting focuses on predicting the relationship between time-series radiation and power data from solar plants with hourly data spanning several years.

Moreover, many studies explore the possibilities and potentials of solar energy usage in cities. Studies of large-scale rooftop solar potential forecasting are conducted at the meso-scale. Typically, such studies identify the segmentation of rooftop boundaries from satellite images and maps, to assess the solar potential at a regional or city scale. Then, the application potentials of PV panels on rooftops are assessed over a large area. The findings from these studies can provide recommendations for the large-scale application potential of PV technology in regions and cities.

2.2. Micro-scale solar energy forecasting

For the micro-scale solar energy forecasting research, the horizontal size is generally less than 2 km, and the available sample size is always less than 5000 with simulated data (Table 1). In some papers, the research on this scale also refers to the neighborhood-scale or block-scale (Lan et al., 2022; Tian and Xu, 2021). The main focus of this research is solar potential forecasting for the existing blocks and parametric models, which both consider the interaction effect of urban environments and buildings (Table 2, Table 3).

Literature focusing on the existing urban environment mainly aims to determine the solar potential by considering the existing morphologies and parameters of neighborhoods, and providing PV recommendations for existing buildings. On the other hand, solar potential forecasting based on the parametric model mainly aims to explore the influence of different layouts and parameter ranges on the solar distribution, and then propose suggestions for optimizing neighborhood layouts and parameters. Generally, the solar potentials of the roof, façade and entire block are evaluated. In contrast to the meso-scale, micro-scale solar forecasting research pays more attention to the details of the urban environments and buildings, including the interaction between multiple buildings, building density, morphology, and orientation.

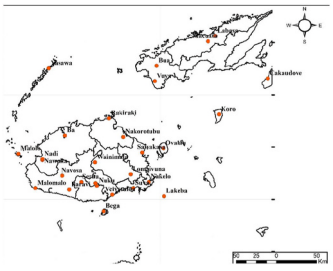
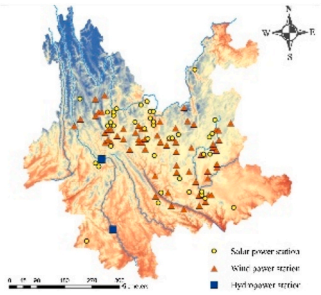
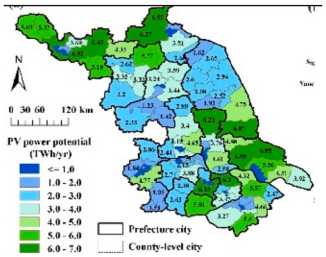
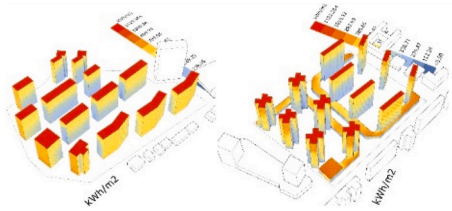
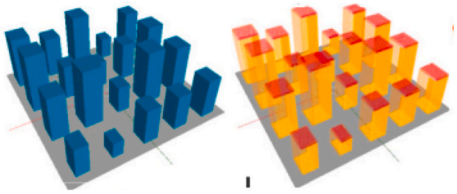
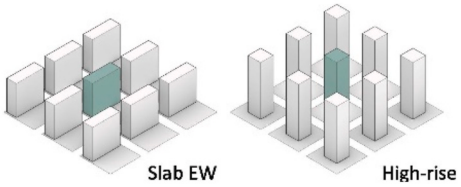
2.3. Building-scale solar energy forecasting

For the building-scale solar energy forecasting research, the horizontal size is generally less than 100 m and the available sample size is always less than 5000 (Table 1). Therefore, studies at this scale primarily rely on the simulated data and parameterized models. From a building-scale perspective, solar energy forecasting is meaningful for assessing the solar potential of an individual building in the context of

Table 1
Multi-scale solar radiation and photovoltaic power forecasting research with scale, size, and database.

Multi-scale Solar radiation and photovoltaic power Forecasting Research									
Research scale	Sample size and main prediction target							Size	Database
	>100	>1000	>2000	>5000	>10,000	>20,000	>50,000		
Meso-scale	Overall solar radiation and solar power prediction, Large-scale rooftop solar potential prediction							<200 km	Experiential Data
Micro-scale	Solar potential prediction for existing urban block or parametric model							<2 km	Simulation/Parameterized
Building-scale	Solar energy prediction for building							<100 m	Simulation/Parameterized

Table 2
Multi-scale solar radiation and photovoltaic power forecasting research with detailed division.

Research Scale	Multi-scale solar radiation and photovoltaic power forecasting		
Meso-scale	Spatial-temporal solar radiation forecasting	Spatial-temporal solar power forecasting	Large-scale rooftop solar photovoltaic potential forecasting
	<200 km		
			
	• Meteorological station (Oyewola et al., 2022)	• Solar power plant (Zhang et al., 2022)	—— Urban boundary (Jiang et al., 2022)
Micro-scale	Solar potential forecasting with the existing urban environment	Solar potential forecasting with the parametric model and optimization	
	<2 km		
			
	Tian and Xu (2021)	Bushra (2022)	
Building-scale	Building-scale solar energy forecasting		
	<100 m		
			
	Natanian et al. (2019)		

surrounding buildings and environments, considering factors such as the building height, interval, shape, and volume (Table 2, Table 3). In addition, both the rooftop and façade solar potentials are evaluated. At this scale, parametric models with simulation processes are commonly employed to derive optimization strategies and recommendations for building design.

3. Machine learning for solar energy forecasting

3.1. Input data and target data

Multiscale solar radiation and photovoltaic power forecasting involves selecting different input data based on the target values (Table 4). Correlation analysis has been conducted in many studies to identify the parameters with significant influence on the target value, which are subsequently used as input parameters for the prediction model. Table 4 lists the diverse input value options for multiple scales.

For the meso-scale solar energy forecasting research, there is a diverse selection of input and target data. In the literature on spatial-temporal solar radiation forecasting, the input value usually contains the time-series data with meteorological parameters, geographical parameters and previous solar radiation values. Among these, meteorological parameters have the most direct relationship with solar radiation, including temperature, humidity, sunshine duration, and wind speed. It is worth noting that the solar radiation directly affects PV power generation. Hence, in the literature on spatial-temporal solar

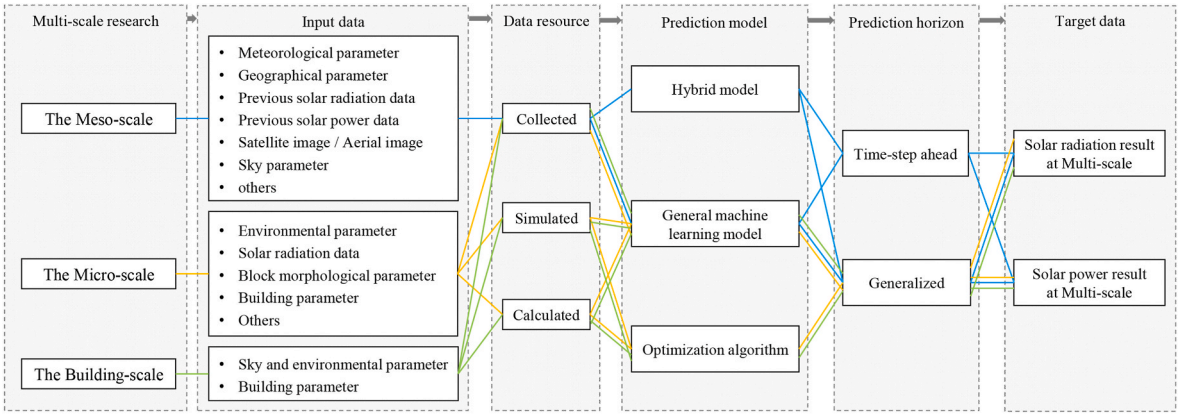


Fig. 2. Multi-scale solar radiation and photovoltaic power forecasting research with similar internal logic.

power forecasting, the input value usually contains time-series data with the solar radiation, previous solar power, and meteorological parameters. In the literature on large-scale rooftop solar potential forecasting, the author mainly focused on the distribution of solar radiation on building rooftops, resulting in unique input requirements. The satellite images, together with many building morphology and geometry parameters, are generally adopted as input values for the neural networks. The CNN and other technologies are usually adopted in this type of literature to recognize the rooftop boundary, and then assess the rooftop solar potential in an urban environment.

The micro-scale solar energy forecasting research concentrates on the solar radiation potential, solar power potential, and economic potential of buildings in a block environment. The input value mainly includes environmental parameters, block morphological parameters, and building parameters. Among them, the environmental parameters (i.e., the sky and weather factors, etc.), the block morphological parameters (i.e., the building layouts and shape factors, etc.), and the building parameters (i.e., the building density, floor area ratio, height, interval, volume, and orientation, etc.) are included. In the literature on solar potential forecasting for the existing urban environment, these input parameters are derived from existing neighborhoods to build the block model. The purpose is to predict the solar potential of existing urban blocks. In the literature on solar potential forecasting with parametric models, these input parameters are set into a certain range to automatically generate multiple parameterized models. The goal is to provide suggestions for maximizing solar potential with optimal parameters.

For the building-scale solar energy forecasting research, the inputs are related to the individual building and surrounding environments, which include the sky, environmental, and building geometry parameters. Parameters such as building length, width, height, volume, orientation, and the interval between surrounding buildings, are adjusted within specific ranges to generate multiple parameterized building models.

3.2. Data resource

Machine learning and deep learning models are training and learning based on large datasets to make predictions for the targets. In the literature on multi-scale solar radiation and photovoltaic power forecasting, the data resources show obvious characteristics, including the collected data, the simulated data from software, and the calculated data from the polynomial (Table 5).

(1) Collected data

For the meso-scale solar energy forecasting research, data resources are mainly collected from the experiential data. In the literature on spatial-temporal solar radiation forecasting, the dataset is generally collected from meteorological stations with hourly data spanning several years. Many studies have collected data from several meteorological stations in a district to predict the solar radiation in different climate zones. For spatial-temporal solar power forecasting, the dataset is generally collected from the PV systems or solar power plants with several years of real-time hourly data. In the literature on large-scale rooftop solar photovoltaic potential forecasting, satellite images or aerial images are usually collected from maps. These images are subsequently fed into the CNN model to extract the features and segmentations of rooftop recognition. Generally, the solar radiation of a district is collected to assess the rooftop's solar potential. Some studies use technologies, such as LiDAR technology and GIS-based tools, to directly obtain the building 3D models, rooftop parameters, and solar radiation values.

The data processing is performed based on the collected datasets. Standardization and normalization are two general ways to rescale data. They can avoid unstable conditions, gradient explosion and other phenomena in the learning process. Standardization rescales a dataset to have a mean of 0 and a standard deviation of 1, whereas normalization rescales a dataset such that each value falls between 0 and 1. A dataset of images is generally processed to a certain image size to facilitate the training process.

(2) Simulated data from software

For the micro-scale and building-scale solar energy forecasting, block and building parameters are usually collected and set to build a 3D model. These 3D models of real blocks and parameterized models serve as the basis for using simulation software to simulate the solar radiation of the building surface. Grasshopper plugins, including the Ladybug and Honeybee, are mostly used to calculate solar radiation over a yearly or monthly period. The ArcGIS is also utilized in some studies. For the parameterized models in some studies, the written Python program and an open-source Python Library are connected to the simulation software. This connection allows for the optimization of urban and building design parameters by forecasting solar potential.

(3) Calculated data from polynomial

In the literature, the calculated data from a polynomial play a vital role within dataset. These data provide additional possibilities and avenues for exploration. The calculated data usually fall into two categories: the calculated data based on the prediction results, and the calculated data before the prediction process.

The calculated data based on the prediction results can lead to more malleable and diverse results. For the large-scale rooftop solar photovoltaic potential forecasting, the calculations are mainly based on the

Table 3
Index of papers for multi-scale solar radiation and photovoltaic power forecasting research.

Research Scale	Multi-scale solar energy forecasting	Index of paper
Mesoscale	Spatial-temporal solar radiation forecasting	(Yadav and Chandel, 2015; Yadav et al., 2015; Mohammadi et al., 2015; Alsina et al., 2016; Mensour et al., 2017; Khosravi et al., 2018; Kaba et al., 2018; Zhao et al., 2019; Mohanty et al., 2019; Li et al., 2019; Benali et al., 2019; Walch et al., 2019a; Fan et al., 2020a, 2020b; Feng et al., 2020; Marzouq et al., 2020; Pang et al., 2020; Lai et al., 2021; Zhou et al., 2021; Puah et al., 2021; Bellido-Jiménez et al., 2021; Narvaez et al., 2021; Husain and Khan, 2021; Guermoui et al., 2022; Wen et al., 2022; Ghimire et al., 2022a, 2022b; Haider et al., 2022; Jia et al., 2022; Oyewola et al., 2022; Nourani et al., 2022; Geetha et al., 2022; Goliatt and Yaseen, 2023)
	Spatial-temporal solar power forecasting	(Hiyama and Kitabayashi, 1997; Almonacid et al., 2014; Wolff et al., 2016) (H. Wang et al., 2017; Cervone et al., 2017; Kumar and Kalavathi, 2018; Wang et al., 2019; Gao et al., 2019) (F. Wang et al., 2020a; Li et al., 2020; Zang et al., 2020; Theocharides et al., 2020; Kong et al., 2020) (F. Wang et al., 2022a; Shams et al., 2021; Guermoui et al., 2021; Jebli et al., 2021; Chang et al., 2021) (L. Wang et al., 2022c; Etxegarai et al., 2022; Agga et al., 2022) (X. Wang et al., 2022f) (J. Wang et al., 2022b; Zhang et al., 2022; Rai et al., 2023)
	Large-scale rooftop solar photovoltaic potential forecasting	(Mainzer et al., 2017; Assouline et al., 2017; Mohajeri et al., 2018; Mansouri Kouhestani et al., 2019; Castello et al., 2019; Huang et al., 2019; Pinna and Massidda, 2020; Martín-Jiménez et al., 2020; Tiwari et al., 2020; Swierc et al., 2020; Walch et al., 2020; Zhong et al., 2021; Sun et al., 2022; Jiang et al., 2022)
Micro-scale	Solar potential forecasting with the existing urban environment	(Mavsar et al., 2019; Zhou et al., 2022; Lan et al., 2021, 2022; Hosney Lila et al., 2022; Tian and Xu, 2021; Xu et al., 2021; Ren et al., 2022; Veisi and Shakibamanesh, 2022)
	Solar potential forecasting with the parametric model and optimization	(Vartholomaios, 2019; Shakibamanesh and Veisi, 2022; Veisi et al., 2022b)
Building-scale	Building-scale solar energy forecasting	(Natanian et al., 2019; Zhang et al., 2019; Fuentes et al., 2020; Poon et al., 2020; Chen et al., 2021; Krapf et al., 2021; Shi et al., 2021) (J. Wen et al., 2022b) (J. Wen et al., 2022a)

rooftop segmentation results from the forecasting model. Different installation approaches for PV panels are adopted to calculate the available roof area for PV installation on the flat and pitched rooftop segmentations. It can further represent the roof PV power potential. The other condition is the calculated data based on the simulated results before the prediction process, which are then inserted into the prediction model as the target data. For the micro-scale and building-scale solar energy forecasting research, the simulation results of solar radiation for blocks and building surfaces can be directly considered as the solar potential. Some studies use the solar radiation values to calculate the PV power, economic payback period, etc. These calculations are based on polynomials influenced by a few factors, including photovoltaic promotion policies in different localities.

3.3. Machine learning model

The essence of artificial intelligence (AI), which has made breakthroughs in recent years, is the hope that machines can think like human brains (Ciulla et al., 2019). Machine learning (ML) is a critical branch of AI that has witnessed rapid growth. ML aims to learn patterns from vast datasets, and then use these patterns to make predictions regarding unknown data. Among ML algorithms, deep learning (DL) uses a deep neural network structure to train and predict data. Both ML and DL are widely used in the multiscale solar radiation and photovoltaic power forecasting research, as shown in Table 6. It can be seen from Table 6 that ANN model is the most widely used deep learning model, followed by the CNN and LSTM models. The SVM model is the most employed machine learning model, followed by the RF model. Moreover, hybrid models are frequently utilized in many literatures.

3.3.1. Deep learning model

Deep learning utilizes various algorithms based on different network structures, including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short Term Memory networks (LSTM), and Gated Recurrent Units (GRU).

An ANN is based on a collection of connected units or nodes, which are computing systems inspired by biological neural networks that constitute animal brains (Veisi et al., 2022b). The ANN structure generally contains three layers: the input layer, the hidden layer, and the output layer. Each neuron in a single layer connects to all neurons in the previous and next layers, creating a fully connected ANN structure that can transmit a signal to other neurons. The output of each neuron is computed by a nonlinear function of the sum of its inputs, with the weight adjusted as learning proceeds.

The CNN is a class of neural networks commonly applied to analyze visual imagery (Valueva et al., 2020). A CNN consists of different layers, including the input, convolutional, pooling, flattened, fully connected, and output layers (Zang et al., 2020). A distinguishing feature of a CNN is that many neurons share the same filter, which contains a vector of weights and biases. The CNN can deal with not only the image data, which represents 2-D data, but also the time-series data, which can be thought as 1-D grid samples with time intervals (Wang et al., 2019). In addition, the CNN-based U-Net network is a fully convolutional network developed specifically for the image segmentation. It can work with very few training images and yield more precise segmentations, as created by Olaf Ronneberger et al., in 2015 (Ronneberger et al., 2015). In the large-scale rooftop solar potential forecasting research, U-Net is utilized to recognize rooftop segmentations using satellite images for solar potential assessment.

The RNN is a class of neural networks in which connections between nodes can create a cycle. The key point of an RNN is to deal with sequential or time-series data, which can process variable-length sequences of inputs with their internal memory (Kumari and Toshniwal, 2021). The distinguishing feature of RNN is that its hidden layer at t time step is affected not only by the input value of the current step, but also by the calculated weight of the previous step. Therefore, this makes the output of each task keeps a certain amount of memory from the previously stored information. However, the RNN still suffers some limitations such as the gradient vanishing and exploding. Hence, two neural networks derived from the RNN are proposed, LSTM and GRU model, which can alleviate the limitation with unique gates.

The LSTM network, developed by Sepp Hochreiter and Jürgen Schmidhuber in 1997, is an improved model of the RNN and has excellent performance in time-series prediction (Hochreiter and Schmidhuber, 1997). The LSTM network aims to provide the short-term

Table 4
Index of papers for inputs and outputs in multi-scale solar forecasting research.

Research Scale	Multi-scale solar energy forecasting	Input data	Target data	Index of paper
Meso-scale	Spatial-temporal solar radiation forecasting	Meteorological parameter	Solar radiation	(Yadav and Chandel, 2015; Yadav et al., 2015; Mohammadi et al., 2015; Alsina et al., 2016; Mensour et al., 2017; Khosravi et al., 2018; Kaba et al., 2018; Mohanty et al., 2019; Li et al., 2019; Walch et al., 2019a,b; Fan et al., 2020b; Feng et al., 2020; Fan et al., 2020a; Pang et al., 2020; Lai et al., 2021; Zhou et al., 2021; Bellido-Jiménez et al., 2021; Narvaez et al., 2021; Husain and Khan, 2021; Guermoui et al., 2022; Ghimire et al., 2022b; Haider et al., 2022; Jia et al., 2022; Ghimire et al., 2022a; Oyewola et al., 2022; Nourani et al., 2022; Geetha et al., 2022; Goliatt and Yaseen, 2023)
		Geographical parameter		(Yadav and Chandel, 2015; Yadav et al., 2015; Mohammadi et al., 2015; Alsina et al., 2016; Mensour et al., 2017; Khosravi et al., 2018; Walch et al., 2019a,b; Nourani et al., 2022)
		Previous solar radiation data		(Zhao et al., 2019; Benali et al., 2019; Marzouq et al., 2020; Lai et al., 2021; Puah et al., 2021; Narvaez et al., 2021; Guermoui et al., 2022) (H. Wen et al., 2022; Haider et al., 2022; Nourani et al., 2022)
		others		(Zhao et al., 2019; Fan et al., 2020b; Lai et al., 2021; Bellido-Jiménez et al., 2021; Jia et al., 2022)
	Spatial-temporal solar power forecasting	Meteorological parameter	Solar power	(Hiyama and Kitabayashi, 1997; Almonacid et al., 2014; Wolff et al., 2016; Cervone et al., 2017; Kumar and Kalavathi, 2018; Wang et al., 2019; Gao et al., 2019; Li et al., 2020; Zang et al., 2020; Theocharides et al., 2020) (F. Wang et al., 2022a; Jebli et al., 2021; Chang et al., 2021) (L. Wang et al., 2022c; Etxegarai et al., 2022; Agga et al., 2022) (X. Wang et al., 2022f) (J. Wang et al., 2022b; Zhang et al., 2022)
		Solar radiation data		(Wolff et al., 2016; Wang et al., 2017; Gao et al., 2019; Li et al., 2020; Zang et al., 2020; Theocharides et al., 2020; Jebli et al., 2021; Chang et al., 2021; Etxegarai et al., 2022) (X. Wang et al., 2022f) (J. Wang et al., 2022b; Zhang et al., 2022; Rai et al., 2023)
		Previous solar power data		(Wolff et al., 2016; Wang et al., 2017; Gao et al., 2019) (F. Wang et al., 2020a; Li et al., 2020; Zang et al., 2020; Kong et al., 2020; Rana and Rahman, 2020) (F. Wang et al., 2022a; Guermoui et al., 2021) (L. Wang et al., 2022c; Agga et al., 2022)
		others		(Cervone et al., 2017; Gao et al., 2019; Zang et al., 2020; Theocharides et al., 2020; Kong et al., 2020) (F. Wang et al., 2022a; Shams et al., 2021; Agga et al., 2022)
	Large-scale rooftop solar photovoltaic potential forecasting	Meteorological parameter	Roof segmentation, rooftop solar radiation potential and solar power potential	(Mainzer et al., 2017; Assouline et al., 2017; Mohajeri et al., 2018; Mansouri Kouhestani et al., 2019; Pinna and Massidda, 2020; Swierc et al., 2020; Walch et al., 2020; Jiang et al., 2022)
		Satellite image/ Aerial image		(Mainzer et al., 2017; Castello et al., 2019; Huang et al., 2019; Martín-Jiménez et al., 2020; Tiwari et al., 2020; Zhong et al., 2021; Sun et al., 2022; Jiang et al., 2022)
		Solar radiation data		(Mainzer et al., 2017; Mohajeri et al., 2018; Mansouri Kouhestani et al., 2019; Pinna and Massidda, 2020; Swierc et al., 2020; Walch et al., 2020; Jiang et al., 2022)
		Building parameter and others		(Mainzer et al., 2017; Assouline et al., 2017; Mohajeri et al., 2018; Mansouri Kouhestani et al., 2019; Pinna and Massidda, 2020; Martín-Jiménez et al., 2020; Tiwari et al., 2020; Swierc et al., 2020; Walch et al., 2020; Sun et al., 2022; Jiang et al., 2022)
Micro-scale	Solar potential forecasting with the existing urban environment and parametric model	Environmental parameter	Solar radiation potential, solar power potential, solar economic potential	(Mavsar et al., 2019; Lan et al., 2021; Ren et al., 2022; Veisi and Shakibamanesh, 2022; Vartholomaios, 2019)
		Block morphological parameter		(Hosney Lila et al., 2022; Tian and Xu, 2021; Xu et al., 2021; Lan et al., 2022; Shakibamanesh and Veisi, 2022; Veisi et al., 2022)
		Building parameter and others		(Mavsar et al., 2019; Zhou et al., 2022; Lan et al., 2021, 2022; Hosney Lila et al., 2022; Tian and Xu, 2021; Xu et al., 2021; Ren et al., 2022; Veisi and Shakibamanesh, 2022; Vartholomaios, 2019; Shakibamanesh and Veisi, 2022; Veisi et al., 2022)
Building-scale	Building-scale solar energy forecasting	Sky and environmental parameter	Building solar radiation potential, Building solar power potential	(Natanian et al., 2019; Poon et al., 2020; Chen et al., 2021)
		Building parameter		(Natanian et al., 2019; Zhang et al., 2019; Fuentes et al., 2020; Poon et al., 2020; Chen et al., 2021; Krapf et al., 2021; Shi et al., 2021) (J. Wen et al., 2022b) (J. Wen et al., 2022a)

memory that can last thousands of time steps, thus "long short-term memory." An LSTM unit is composed of a cell, an input gate, an output gate, and a forget gate (Gers et al., 1999; L. Wang et al., 2022c). The cell retains information over arbitrary time intervals, and the three gates regulate the flow of information into and out of the cell. The LSTM network is a unit with a memory cell, the state of which contains the previously entered information.

The GRU network, introduced in 2014 by Cho et al. is a powerful learning model for sequential data (Cho et al., 2014). Like the LSTM, the GRU employs a gating mechanism to handle the flow of sequence

Table 5
Index of papers for different data resource in multi-scale solar forecasting research.

Research Scale	Multi-scale solar energy forecasting	Index of paper		
		Collected data	Simulated data from software	Calculated data from polynomial
Meso-scale	Spatial-temporal solar radiation forecasting	(Yadav and Chandel, 2015; Yadav et al., 2015; Mohammadi et al., 2015; Alsina et al., 2016; Mensour et al., 2017; Khosravi et al., 2018; Kaba et al., 2018; Zhao et al., 2019; Mohanty et al., 2019; Li et al., 2019; Benali et al., 2019; Walch et al., 2019a; Fan et al., 2020a, 2020b; Feng et al., 2020; Marzouq et al., 2020; Pang et al., 2020; Lai et al., 2021; Zhou et al., 2021; Puah et al., 2021; Bellido-Jiménez et al., 2021; Narvaez et al., 2021; Husain and Khan, 2021; Guermoui et al., 2022) (H. Wen et al., 2022; Ghimire et al., 2022b; Haider et al., 2022; Jia et al., 2022; Ghimire et al., 2022a; Oyewola et al., 2022; Nourani et al., 2022; Geetha et al., 2022; Goliatt and Yaseen, 2023)	Marzouq et al. (2020)	(Benali et al., 2019; Feng et al., 2020)
	Spatial-temporal solar power forecasting	(Hiyama and Kitabayashi, 1997; Almonacid et al., 2014; Wolff et al., 2016; Wang et al., 2017, 2019; Cervone et al., 2017; Kumar and Kalavathi, 2018; Gao et al., 2019) (F. Wang et al., 2020a; Li et al., 2020; Zang et al., 2020; Theocharides et al., 2020; Kong et al., 2020; Rana and Rahman, 2020) (F. Wang et al., 2022a; Shams et al., 2021; Guermoui et al., 2021; Jebli et al., 2021; Chang et al., 2021) (L. Wang et al., 2022c; Etxegarai et al., 2022; Agga et al., 2022) (X. Wang et al., 2022f) (J. Wang et al., 2022b; Zhang et al., 2022; Rai et al., 2023)	(F. Wang et al., 2022a)	(Wolff et al., 2016; Etxegarai et al., 2022; Zhang et al., 2022)
	Large-scale rooftop solar photovoltaic potential forecasting	(Mainzer et al., 2017; Assouline et al., 2017; Mohajeri et al., 2018; Mansouri Kouhestani et al., 2019; Castello et al., 2019; Huang et al., 2019; Pinna and Massidda, 2020; Martín-Jiménez et al., 2020; Tiwari et al., 2020; Swierc et al., 2020; Walch et al., 2020; Zhong et al., 2021; Sun et al., 2022; Jiang et al., 2022)		
Micro-scale	Solar potential forecasting with the existing urban environment	(Mavsar et al., 2019; Zhou et al., 2022; Lan et al., 2021, 2022; Hosney Lila et al., 2022; Tian and Xu, 2021; Xu et al., 2021; Ren et al., 2022; Veisi and Shakibamanesh, 2022)		
	Solar potential forecasting with the parametric model	(Vartholomaios, 2019; Shakibamanesh and Veisi, 2022; Veisi et al., 2022b)		
Building-scale	Building-scale solar energy forecasting	(Natanian et al., 2019; Zhang et al., 2019; Fuentes et al., 2020; Poon et al., 2020; Chen et al., 2021; Krapf et al., 2021; Shi et al., 2021) (J. Wen et al., 2022b) (J. Wen et al., 2022a)		

information. However, the GRU network contains only two gates: an update gate and a reset gate (Niu et al., 2023). These two gates can help the network determine the useful information that should be retained from the past, and irrelevant information that should be removed (Kumari and Toshniwal, 2021; Niu et al., 2023).

3.3.2. Other machine learning models

Machine learning is usually divided into two problems: classification and regression. Common algorithms used for these tasks include the Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Extreme Gradient Boost (XGBoost).

The RF is an ensemble learning method that operates by training a multitude of decision trees and grouping their results. The RF forecasts the output value of each decision tree by recognizing and learning the rules and patterns from the input feature (Rana and Rahman, 2020). In classification tasks, the RF outputs the different groups selected by the most trees. In regression tasks, the output is the average prediction of the individual trees (Ho, 1998).

The SVM is a supervised learning algorithm. The core of an SVM is the ability to map the input data into a higher-dimensional feature space to recognize the rules between the feature and target values. In classification tasks, the SVM trains examples to points in a space to find the optimal hyperplane that maximizes the interval between the points of each category, and uses it to classify them in the new space. In regression tasks, the Support Vector Regression (SVR) algorithm applies the linear regression for the nonlinear mapping of the input time-series data in a large feature space. The SVM is widely applied for solar forecasting because it is less likely to overfit (Ağbulut et al., 2021; Jebli et al., 2021; Jia et al., 2022; Rana and Rahman, 2020; Shams et al., 2021).

The KNN is a classical supervised learning method (Chen et al., 2013). In the KNN classification, a special number k is defined, and k samples of the nearest neighbors close to the input object are obtained in the feature space. If most of these k-nearest neighbors belong to a particular class, the input value is identified as belonging to this class. In the KNN regression, the average value of these neighbors is assigned to the input object by finding the k-nearest neighbors of the input sample (Ağbulut et al., 2021). The KNN algorithm is simple, effective, and suitable for samples with large feature space dimensions and large data capacity (Piryonesi and El-Diraby, 2020).

The XGBoost is a supervised algorithm for the tree boosting that has been widely recognized in many data mining challenges (Chen and Guestrin, 2016). The XGBoost uses fixed-size decision trees to develop a prediction model. It uses a theoretically justified weighted quantile sketch procedure that enables the handling of instance weights in tree learning (Chen and Guestrin, 2016). Furthermore, the XGBoost incorporates the parallel and distributed computations during the training process, which enables a faster model exploration and improved computational efficiency (Ghimire et al., 2022b; Goliatt and Yaseen, 2023).

3.3.3. Hybrid model

In addition to using a separate forecasting model, the hybrid model has also been adopted in the solar forecasting research to deal with more complex conditions and improve forecasting performance. In the literature, the hybrid model with multiple machine learning structures and optimization algorithms can be generally considered an encoder-decoder model. It has two parts: an encoder can read and extract data features from the input sequence, and a decoder can read the encoded sequence and make predictions for the output sequence (Agga et al., 2022).

The hybrid model of the CNN-LSTM architecture is widely used and has been verified to have an excellent performance in forecasting time-series datasets. The CNN can extract spatial features, whereas the LSTM can extract temporal features (Wang et al., 2019). The CNN layer extracts the unseen data features in the time series after preprocessing, and create vectors used as inputs for the decoding model. The LSTM

Table 6
Index of papers for prediction models in multi-scale solar forecasting research.

Machine learning model		Index of paper
Deep learning models	Artificial Neural Networks (ANN)	(Yadav and Chandel, 2015; Yadav et al., 2015; Alsina et al., 2016; Mensour et al., 2017; Mohanty et al., 2019; Li et al., 2019, 2020; Benali et al., 2019; Pang et al., 2020; Lai et al., 2021; Puah et al., 2021; Bellido-Jiménez et al., 2021; Narvaez et al., 2021; Husain and Khan, 2021; Guermoui et al., 2022; Ghimire et al., 2022b; Haider et al., 2022; Oyewola et al., 2022; Geetha et al., 2022; Hiyama and Kitabayashi, 1997; Almonacid et al., 2014; Cervone et al., 2017; Kumar and Kalavathi, 2018; Zang et al., 2020; Theocharides et al., 2020; Rana and Rahman, 2020; Shams et al., 2021; Jebli et al., 2021; Etxegarai et al., 2022; Agga et al., 2022) (X. Wang et al., 2022f; Zhang et al., 2022; Swierc et al., 2020; Walch et al., 2020; Jiang et al., 2022; Lan et al., 2021; Hosney Lila et al., 2022; Lan et al., 2022; Veisi et al., 2022b; Chen et al., 2021)
	Convolutional Neural Networks (CNN)	(Zhao et al., 2019; Guermoui et al., 2022; Haider et al., 2022; Ghimire et al., 2022a; Wang et al., 2017, 2019; Zang et al., 2020; Kong et al., 2020; Chang et al., 2021; Etxegarai et al., 2022; Agga et al., 2022) (J. Wang et al., 2022b; Rai et al., 2023; Mainzer et al., 2017; Castello et al., 2019; Huang et al., 2019; Zhong et al., 2021; Sun et al., 2022; Jiang et al., 2022; Ren et al., 2022; Krapf et al., 2021)
	Recurrent Neural Networks (RNN)	(Pang et al., 2020) (F. Wang et al., 2020a; Li et al., 2020; Zhang et al., 2022; Rai et al., 2023)
	Long Short Term Memory networks (LSTM)	(Lai et al., 2021; Narvaez et al., 2021; Guermoui et al., 2022) (H. Wen et al., 2022; Haider et al., 2022; Ghimire et al., 2022a; Nourani et al., 2022; Wang et al., 2019; Gao et al., 2019) (F. Wang et al., 2020a; Li et al., 2020; Kong et al., 2020; Shams et al., 2021; Chang et al., 2021) (L. Wang et al., 2022c; Etxegarai et al., 2022; Agga et al., 2022) (J. Wang et al., 2022b; Rai et al., 2023)
	Gated Recurrent Unit (GRU)	(Narvaez et al., 2021) (H. Wen et al., 2022; Li et al., 2020) (L. Wang et al., 2022c; Rai et al., 2023)
Other machine learning models	Random Forest (RF)	(Benali et al., 2019; Fan et al., 2020a; Marzouq et al., 2020; Bellido-Jiménez et al., 2021; Narvaez et al., 2021; Husain and Khan, 2021; Jia et al., 2022; Ghimire et al., 2022a; Zang et al., 2020; Rana and Rahman, 2020; Shams et al., 2021; Jebli et al., 2021) (X. Wang et al., 2022f; Swierc et al., 2020; Walch et al., 2020; Lan et al., 2022)
	Support Vector Machine (SVM)	(Mohammadi et al., 2015; Mohanty et al., 2019; Fan et al., 2020a, 2020b; Feng et al., 2020; Bellido-Jiménez et al., 2021; Husain and Khan, 2021; Guermoui et al., 2022; Jia et al., 2022; Ghimire et al., 2022a; Wolff et al., 2016; Wang et al., 2017; Gao et al., 2019) (F. Wang et al., 2020a; Zang et al., 2020; Rana and Rahman, 2020) (F. Wang et al., 2022a; Shams et al., 2021; Jebli et al., 2021) (L. Wang et al., 2022c) (X. Wang et al., 2022f; Zhang et al., 2022; Assouline et al., 2017; Mohajeri et al., 2018; Swierc et al., 2020; Vartholomaios, 2019)
	K-Nearest Neighbors (KNN)	(Husain and Khan, 2021) (L. Wang et al., 2022c; Agga et al., 2022; Swierc et al., 2020; Lan et al., 2022)
	Extreme Gradient Boost (XGBoost)	(Fan et al., 2020a, 2020b; Bellido-Jiménez et al., 2021; Goliatt and Yaseen, 2023)
Hybrid model		(Khosravi et al., 2018; Zhao et al., 2019; Mohanty et al., 2019; Walch et al., 2019a; Fan et al., 2020b; Feng et al., 2020; Marzouq et al., 2020; Lai et al., 2021; Zhou et al., 2021; Puah et al., 2021; Guermoui et al., 2022) (H. Wen et al., 2022; Ghimire et al., 2022b; Ghimire et al., 2022a; Nourani et al., 2022; Goliatt and Yaseen, 2023; Wang et al., 2017; Cervone et al., 2017; Wang et al., 2019) (F. Wang et al., 2020a; Li et al., 2020; Zang et al., 2020) (F. Wang et al., 2022a; Guermoui et al., 2021) (L. Wang et al., 2022c; Agga et al., 2022) (J. Wang et al., 2022b; Rai et al., 2023; Ren et al., 2022)

architecture receives the extracted features from the encoder as inputs to discover the connections within the input and output sequences (Agga et al., 2022).

3.4. Forecasting horizon

The solar radiation and power values have a strong relationship with the timescale. Overall, the summer period has the highest solar radiation of one year, and the time around midday has a high solar radiation of one day, directly affecting the solar power generation. Therefore, in the multiscale solar forecasting research, various forecasting horizons are utilized based on time-series datasets.

Generally, the forecasting horizon will be set as the minutely, hourly, daily, monthly, seasonal, and annual horizon. Depending on the timescales of the forecasting horizon, the solar radiation and power prediction can be categorized as the generalized prediction and time-step-ahead prediction (Fig. 3).

(1) Generalized prediction

Generalized solar energy prediction usually refers to the prediction of daily, monthly, seasonal, and annual horizons. This type of prediction has the one-to-one corresponding relations between inputs and outputs, which are quite common in neural network models for multiscale solar energy forecasting (Fig. 3 a). The input value consists of several parameters, and the target value is generally solar radiation or PV power that can correspond to each set of input values. The literature usually screens input values through correlation analysis before the prediction, and some make different input value combinations in the prediction process to verify optimal forecasting conditions.

(2) Time-step-ahead prediction

The time-step-ahead prediction is mostly used in the prediction for minutely, hourly, daily, and monthly horizons, using detailed time-series datasets. The past outputs within a certain timescale are used as the inputs, along with other relevant input parameters, to forecast the target value for the next timescale (Fig. 3 b). The time-step-ahead prediction focuses on the correlation within the timescale, using multi-step-ahead data to forecast the future tendency of the data. Some studies only consider the past solar power values to predict future solar power, whereas most studies incorporate other input values to co-predict future data.

4. Meso-scale solar energy forecasting

Fig. 4 shows the meso-scale solar energy forecasting research. A general comparison of all literature was carried out for each field, considering the characteristics summarized in Tables 7–12.

- Input and target data: In the literature on multiscale solar energy forecasting, different target values lead to various selections for the input data.
- The data resource includes the collected data, the simulated data from the software, and the calculated data from the polynomial.
- Machine learning model: different selections for machine learning, deep learning, and hybrid models in the literature. Some studies compare many models and provide the best prediction model.

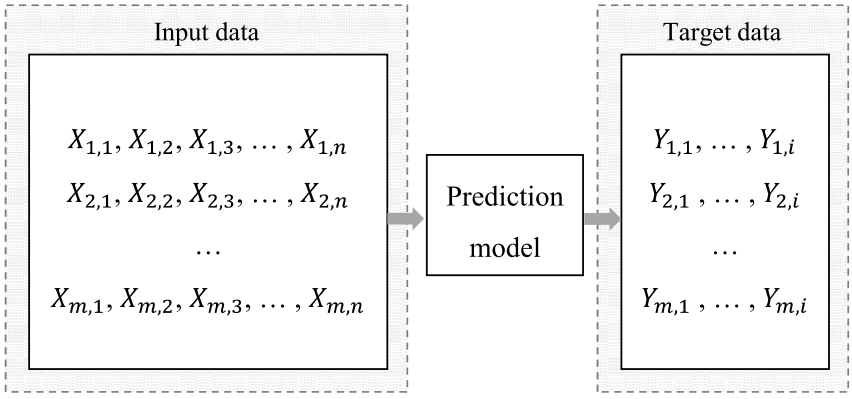
- Forecasting horizon: Two methods for the forecasting horizon are summarized: generalized prediction and time-step-ahead prediction. These methods consider potential influences of weather and seasonal factors.
- Evaluation indicators: Commonly used indicators are analyzed to evaluate the prediction errors.

4.1. Spatial-temporal solar radiation forecasting

Solar radiation forecasting in urban environments is generally carried out using time-series datasets that consist mainly of meteorological and geographical parameters. The literature verifies the prediction performance of different machine learning models (Table 7, Table 8), respectively. Table 7 (i.e., inputs and outputs, data resources, location and period) and Table 8 (i.e., prediction models, best models, evaluation indicators, forecasting horizons) show the information of papers in the spatial-temporal solar radiation forecasting research.

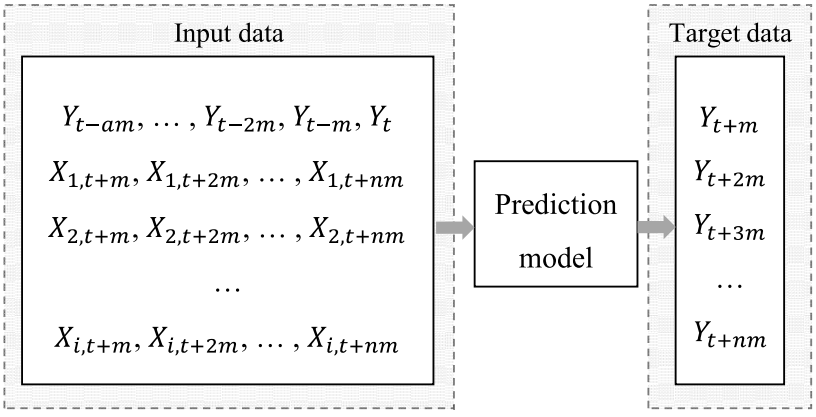
An ANN model with multiple layers is commonly used and has been verified to exhibit good performance for solar radiation forecasting. Yadav et al. (Yadav and Chandel, 2015) assessed the solar potential of western India using an ANN-based global solar radiation prediction model. Subsequently, they (Yadav et al., 2015) proposed a new technique, Rapid Miner, for the selection of relevant input variables for solar radiation prediction using different models: ANN, RBFNN, and GRNN. The comparison showed that ANN models achieved the best prediction with MAPE ranging from 6.89% to 16.91%. Alsina et al. (2016) and Mensour et al. (2017) focused on predicting Global Solar Radiation using Multi-Layer Perceptron (MLP) model. A relevant analysis of the input data was adopted to generate several configurations with MAPE ranging from 1.67% to 4.25%, emphasizing the most critical parameters. Oyewola et al. (2022) and Geetha et al. (2022) developed an ANN model with optimization algorithms to forecast solar radiation using commonly available meteorological and geographical data. Their models achieved an MSE lower than 0.12 and a correlation coefficient of 0.94.

The CNN model can handle both image and time-series data, which are widely used to predict solar sequences. Zhao et al. (2019) proposed a novel 3D-CNN method for predicting the direct normal irradiance (DNI) by processing consecutive cloud images to extract cloud features. A typical linear autoregressive model (AR) and MLP were selected as forecasting models, in which the features from the last fully connected layer in the CNN served as inputs. The MLP coupled with the 3D–CNN–based model had the best forecasting accuracy for clear-sky and broken-sky, achieving improvements of 17.06% and 28.89% in nRMSE over the persistent model, respectively. Kaba et al. (2018) estimated the global solar radiation using a simple DNN model. The results indicated that sunshine hours were the most important input, followed by cloud cover. The input combinations that include sunshine hours



n : Category of input data. i : Category of target data. m : Number of dataset.

(a) Generalized prediction.



t : Time. m : Time-step. a : Number of time-step in past timescale.
i : Category of input data. n : Number of time-step in future timescale.

(b) Time-step-ahead prediction.

Fig. 3. Forecasting horizon.

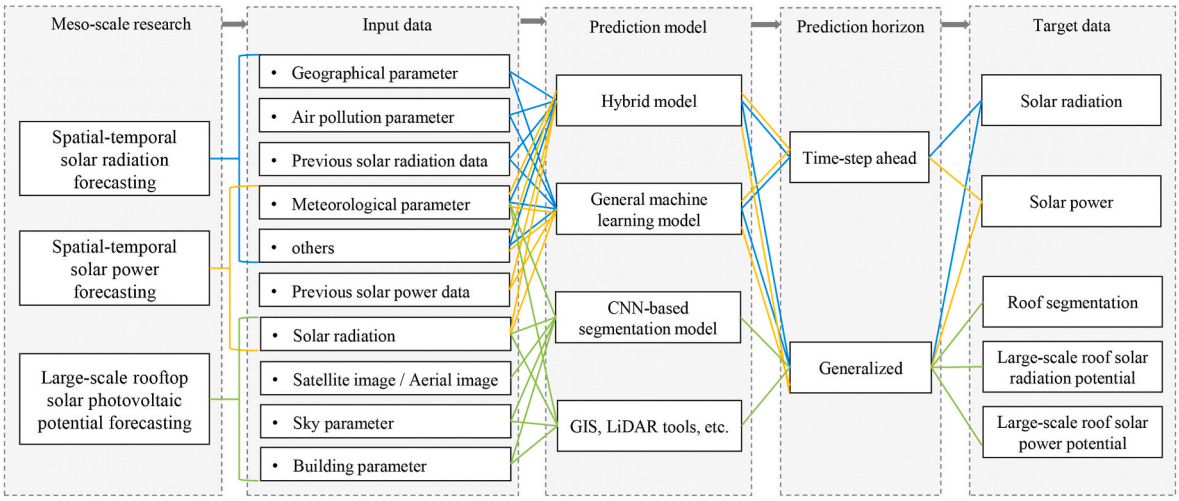


Fig. 4. Meso-scale solar energy forecasting research.

produced relatively lower errors, in which MAE and RMSE changing from 0.61 to 0.7, and 0.78 to 0.91, respectively.

The RNN, LSTM, and GRU models are generally utilized to predict multi-step-ahead sequences. Pang et al. (2020) compared the ANN and RNN models to investigate the performance of solar radiation prediction. The effects of the data sampling frequency and moving window algorithm with different time steps on the prediction performance were analyzed. The RNN model exhibited a relatively high prediction accuracy with a coefficient of variation of RMSE of 7.64, which was improved by 19%. Nourani et al. (2022) proposed multifrequency LSTM methods of sequenced-LSTM and wavelet-LSTM for multi-step-ahead solar irradiance simulation, which can serve all the capabilities of classic LSTM networks and handle their weaknesses in detecting and modeling information. The proposed multifrequency LSTM network can improve the performance of LSTM in the test phase by up to 13%.

General machine learning models with different algorithms, such as SVM, RF, and MARS, have been verified to have excellent forecasting performance. Mohammadi et al. (2015) adopted the SVR model to estimate the horizontal global radiation. They assessed two SVRs, a radial basis function (SVR-rbf) and a polynomial basis function (SVR-poly), to predict both daily and monthly mean scales. The result shows that SVR-rbf demonstrated superior performance for daily estimation with an RMSE of 2.00, and both SVRs showed excellent performance for monthly mean estimation with RMSE of 0.45 and 0.78, respectively. Jia et al. (2022) assessed SVM, RF, and GLMNET to predict the daily global and diffuse solar radiation under different weather and pollution conditions in eight Chinese cities. The results showed that SVM was the best performance model to predict global and diffuse solar radiation with nMAE change from 10.33 to 28.85, 19.73 to 40.42, respectively. Li et al. (2019) developed models for predicting hourly global solar radiation based on the MARS method. The comparison analysis showed that the MARS models performed well in terms of accuracy and interpretability with an RMSE ranging from 0.27 to 0.74.

Most studies have proposed a hybrid model that incorporate optimization algorithms to improve the prediction performance. Fan et al. (2020b) proposed three new hybrid SVMs with a particle swarm optimization algorithm (SVM-PSO), bat algorithm (SVM-BAT), and whale optimization algorithm (SVM-WOA) for predicting the daily diffuse solar radiation in air-polluted regions. Results showed that the SVM-BAT models enhanced the accuracy of prediction with relative decreases in RMSD of 2.9%–5.6%, and models with air-polluted factors like $PM_{2.5}$, PM_{10}, O_3 produced more accurate predictions. Feng et al. (2020) evaluated a hybrid PSO-ELM to predict the daily global solar radiation. The comparison analysis showed that the hybrid model, with easier implementation and fewer required parameters, provided more accurate estimations with an average MAE of 1.576 and an RRMSE of 0.149.

In addition, a hybrid model with multiple machine learning structures can be generally considered as an encoder-decoder model, enhancing the ability to capture the dataset features. Wen et al. (H. Wen et al., 2022) proposed the STANN model in conjunction with the federated learning (FL) technique for multi-horizon solar irradiance forecasting. STANN can learn the local characteristics of each dataset, and FL can further improve forecasting performance by up to 11.2%. Ghimire et al. (2022b) predicted daily global solar radiation by developing a hybrid CXGBRFR framework, which includes CNN, XGBoost with RF Regression, and a Harris Hawks Optimization. Among them, CNN with HHO was used for the initial feature selection, and XGBoost with RFR was used for solar prediction at daily and seasonal timescales. The hybrid model exhibited excellent performance, achieving an RMSE of 2.118 and an MAE of 1.587. Ghimire et al. (2022a) proposed a CSVR model that integrates a CNN with an SVR approach for predicting daily global solar radiation. The CNN model is used to extract local pattern features in the time series at different intervals, and the SVR replaces the fully connected layer of the CNN to predict the daily GSR. The hybrid CSVR can significantly improve RRMSE over other models by more than 15%.

Some studies have demonstrated the benefits of different machine and deep learning models in integrating data from different sources and constructing precise forecasting models. Guermoui et al. (2022) improved the forecasting of 12-h ahead for hourly global solar radiation by relying on the CNN, LS-SVM, and ELM models. Each model performed better than the other models within a specified horizon. LS-SVM provides high forecasting results for the first 7 h ahead, while CNN achieves high forecasting results in the 9 h-ahead. Haider et al. (2022) used two statistical methods (Prophet and SARIMAX) and three DL architectures (ANN, CNN, and LSTM) to predict the global horizontal irradiance. DL models performed best for 1 h and 3 h ahead short-term forecasting with R2 scores above 0.90, whereas statistical models performed best for one-shot long-term forecasting with R2 scores over 0.92.

4.2. *Spatial-temporal solar power forecasting*

Solar power forecasting (Table 9, Table 10) mainly focuses on the regular rule between radiation and power data in urban environments, because solar radiation has a direct impact on the solar power performance of PV panels. Table 9 (i.e., inputs and outputs, data resources, location and period) and Table 10 (i.e., prediction models, best models, evaluation indicators, forecasting horizons) show the information of papers in the spatial-temporal solar power forecasting research.

An ANN model with an optimization algorithm is commonly used

Table 7
Index of papers for input value, output value and data information in spatial-temporal solar radiation forecasting research.

Ref.	Input value				Target value	Data resource			Location and period
	Meteorological parameter	Geographical parameter	Historical solar radiation	Others		Collected	Simulated	Calculated	
Yadav and Chandel (2015)	T, Tmin, Tmax, SH, CI, ER	Lat, Long, H			Global solar radiation (GSR)	•			26 Indian cities from different climatic zones (1986–2000)
Yadav et al. (2015)	T, Tmin, Tmax, SH, CI, ER	Lat, Long, H			Solar radiation (SR)	•			26 sites in India (1986–2000)
Mohammadi et al. (2015)	SH, N				Global solar radiation	•			The city of Isfahan (1985–1991, 1998–2003)
Alsina et al. (2016)	Tave, ADC, ATDR, CSD, CR, RD, HDD, ADD, SH, Time	Lat, Long, H			Global solar radiation	•			45 Italian locations
Mensour et al. (2017)	Time, Tave, SH, RH	Lat, Long			Global solar irradiation.	•			Souss-Massa area (1994–2003)
Khosravi et al. (2018)	Time, Tave, Tmin, Tmax, AP, RH, WS, TAI	Lat, Long			Global solar radiation	•			12 stations in Iran (2016)
Kaba et al. (2018)	ER, SH, CC, Tmin, Tmax				Global solar radiation	•			34 meteorological stations in Turkey (2001–2007)
Zhao et al. (2019)			The measured DNI (t-2, t-1, t), Ics (t+1)	Cloud images (t-2, t-1, t)	Clear-sky index classes (t+1), Direct irradiance (t+1)	•			The GBC images and DNI data (2013–2014)
Mohanty et al. (2019)	P/Po, T/To, S/So, R/Ro				Solar radiation	•			5 years
Li et al. (2019)	EP, SH, V, CC, WS, T, Twet, Tdew, RH, AP				Global solar radiation	•			Hong Kong (2010–2016)
Benali et al. (2019)			experimental solar radiation data (GHI, BNI, DHI): t, t-1, “, t-n		Solar radiation (GHI, BNI, DHI): t + h (h+1, …, h+6)	•		•	Meteorological station in France (3 years)
Walch et al. (2019a)	x-coordinate, y-coordinate, hour	H			Global Horizontal Irradiation	•			Switzerland (2004–2015)
Fan et al. (2020b)	SH, Tmin, Tmax, GSR/ER			PM2.5, PM10, O3	Diffuse solar radiation	•			Beijin (2014–2017)
Feng et al. (2020)	SH, Tmin, Tmax, ER, VPD, Time				Global solar radiation	•		•	7 stations in northern China (1961–2016)
Fan et al. (2020a)	SH, Tmin, Tmax, RH, GRS/ER				Diffuse Horizontal Irradiation	•			15 weather stations in China (1996–2015)
Marzouq et al. (2020)			historical global solar irradiance (t, t-l, t-2l)		Global solar irradiance (t+1, t+2, “, t+6)	•	•		28 Moroccan cities (2005)
Pang et al. (2020)	T, Tdew, RH, WS, WD, CC				Global solar radiation	•			A weather station located in Tuscaloosa (May 22nd-May 29th, 2016)
Lai et al. (2021)	CI, T, RH, WS, WD, AP		Historical GHI(t-1), clear-sky GHI (t+1)	SZA	Global Horizontal Irradiation (t+1)	•			Itupiranga and Ocala (2005–2017)
Zhou et al. (2021)	SR, T, TD, RH				Global and diffuse solar radiation	•			10 stations in China (2003–2010)
Puah et al. (2021)			Historical solar irradiance values f (t-59) to f(t)		Solar irradiance f (t+60)	•			Semenyih (April 2018–June 2018)
Bellido-Jiménez et al. (2021)	Tmin, Tmax, the previous day Tmin and Tmax, the following morning Tmin, TD, Time, ER			Et, the previous day Et	Daily solar radiation	•			9 datasets in different geoclimatic conditions (2000–2018)

(*continued on next page*)

Table 7 (*continued*)

Ref.	Input value				Target value	Data resource			Location and period
	Meteorological parameter	Geographical parameter	Historical solar radiation	Others		Collected	Simulated	Calculated	
Narvaez et al. (2021)	T, WS, SZA, Time		GHI, DHI, DNI (1, 3, 7 previous days)		Global Horizontal Irradiance (next one day)	•			Colombia (2006–2017)
Husain and Khan (2021)	ER, CI, Dco, Df				Diffuse solar radiation	•			North India (3 years)
Guermoui et al. (2022)	T, RH, AP, CI, Time		Solar radiation (1 h-ahead to 12 h-ahead)		Global solar radiation	•			Algiers (2013–2016)
(H. Wen et al., 2022)			CI (5min-ahead to 30min-ahead)		Global horizontal irradiance (Yt+1, …, Yt+6)	•			Folsom (2014–2016) and NREL datasets (1 year)
Ghimire et al. (2022b)	Evap, RH at Tmax, hfss, Rain, RH. wap, Hs				Global solar radiation	•			6 solar farms in Queensland (1950–2006)
Haider et al. (2022)	WS, T, WD, DHI, DNI		GHI (1, 3, 6, 12, 24 h ahead)		Global Horizontal Irradiation (GHI)	•			Meteorological station in Islamabad (2015–2019)
Jia et al. (2022)	SH, maximum SH			AQI, PM2.5, PM10, O3	Global and diffuse solar radiation	•			8 cities in China (2015–2020)
Ghimire et al. (2022a)	Evap, RH at Tmax, RH at Tmin, hfss, Rain, RH, wap, Tmax, H, WS, hfls, T				Global Solar Radiation	•			6 solar farms in Queensland (1950–2006)
Oyewola et al. (2022)	Time, T, RH, P	Lat, Long, H			Solar radiation	•			31 cities in Fiji (1984–2019)
Nourani et al. (2022)	T, RH, AP, WS		Solar irradiance (1 day, 1 year ahead)		Solar irradiance	•			3 stations in Iran and 3 stations in USA (2001–2021)
Geetha et al. (2022)	Time, T, WS	Lat, Long			Solar radiation	•			6 districts in India
Goliatt and Yaseen (2023)	WV, Hmax, Hmin, Tmax, Tmin, VPD, Evap				Solar radiation	•			4 stations in Burkina Faso (1998–2012)

and has been verified to exhibit good performance for solar power forecasting. Almonacid et al. (2014) proposed a new methodology for forecasting the output of a PV generator 1-h ahead using a dynamic ANN, achieving an RMSE of 3.38%. The results showed that once the values of solar radiance and air temperature in the next hour were available, it was possible to predict the PV output in the next hour. Kumar (Kumar and Kalavathi, 2018) explored the use of ANN and ANFIS models to predict PV power output. ANN-based forecast delivered better results, achieving a minimum error percentage of 72.64%. Theocharides et al. (2020) derived a unified methodology for accurate day-ahead PV power forecasts based on a data-driven machine-learning approach. Results showed that ANN with linear regression coefficients improved performance accuracy by 5.7%. Jebli et al. (2021) presented an approach for predicting both real-time and short-term solar energy using meteorological and solar radiation data based on the LR, RF, SVR, and ANN (MLP) models. ANN model, followed by RF, which has good accuracies for real-time predictions with NRMSE of 0.03 and 0.06, respectively.

The CNN, RNN, and LSTM models are generally utilized to deal with time sequences. Furthermore, combinations of these models, such as CNN-LSTM, have been verified to extract deep feature patterns and improve forecasting performance. Gao et al. (2019) proposed a day-ahead power forecasting approach in which ideal and non-ideal weather types were separately discussed. For ideal weather conditions, an LSTM model was proposed with an RMSE of 4.62%. For non-ideal weather conditions, time-series relevance and specific weather type characteristics were considered based on the LSTM and DGM algorithms. F. Wang et al. (F. Wang et al., 2020a) proposed a day-ahead PV power forecasting model that combined an LSTM-RNN, resulting in an improvement of the RMSE from 8.83% to 5.68%. H. Zang et al. (2020) proposed a day-ahead PV power forecasting approach based on two novel deep CNN models, the residual network (ResNet) and dense convolutional network (DenseNet). These models demonstrated higher representation capabilities, achieving an MAE of 0.152 and 0.180, respectively. Agga et al. (2022) and Rai et al. (2023) suggested a CNN-LSTM hybrid architecture for predicting the PV power. The CNN-LSTM showed an accuracy improvement in predictions, owing to the merging of the significant strengths of each model and the better recognition of erratic power generation patterns. The MAPE of the hybrid model can be improved by over 5.55%.

Machine learning models, such as SVR and RF, are commonly combined with various algorithms for PV power forecasting. Wolff et al. (2016) applied the SVR model for short-term PV power forecasting at single-site and regional levels. The PV measurement-based SVR model performed better for short forecast horizons up to 1 h ahead, whereas NWP-based predictions proved to be better on horizons starting at 3 h. Rana et al. (Rana and Rahman, 2020) presented NN, SVR, RF, and MLR models for multi-step-ahead PV power prediction. They used a data resampling technique that enabled machine learning algorithms to compute multi-step-ahead predictions via simple single-step-ahead prediction on the resampled time-series. The proposed approach with RF achieved the best performance, with an MRE of 10.5%. Shams et al. (2021) proposed RT, GBDT, RF, ANN, LSTM, and SVR models to predict wind and solar power curtailment (WSPCs). By employing the HO approach, the RF method outperformed the other models, producing an RMSE of 0.017–0.027.

Table 8
Index of papers for prediction models, evaluation indicators and forecasting horizon in spatial-temporal solar radiation forecasting research.

Ref.	Information of prediction model					Evaluation indicators	Forecasting horizon				
	Hybrid model	Machine learning model	Compa-ration	Prediction model	Best model		Generalized	Multi-step ahead	Weather factor	Seasonal factor	Prediction horizon
(Yadav and Chandel, 2015)		•	•	ANN with J48 algorithm	ANN-4	MAPE, R	•			•	Monthly/ Annual
(Yadav et al., 2015)		•	•	ANN, RBFNN, GRNN	ANN	MAPE, RMSE, R	•				24 h
(Mohammadi et al., 2015)		•	•	SVR-rbf, SVR-poly		RPE, MAPE, MABE, RMSE, RRMSE, R2	•				24 h/ Monthly
(Alsina et al., 2016)		•	•	ANN based MLP		MAPE, NRMSE, MPBE	•		•		24 h
(Mensour et al., 2017)		•		ANN based MLP		RMSE, MAE, R	•				24 h
(Khosravi et al., 2018)	•	•	•	GMDH, MLFFNN, ANFIS, ANFIS-PSO, ANFIS-GA, ANFIS-ACO	GMDH	R2, MSE, RMSE	•		•		24 h
(Kaba et al., 2018)		•		DNN		R2, MAE, RMSE	•		•		24 h
(Zhao et al., 2019)	•	•	•	3D-CNN coupled AR, 3D-CNN coupled MLP	3D-CNN coupled MLP	nRMSE, nMAE, NMBE, MAPE		•	•		10min
(Mohanty et al., 2019)	•	•	•	ANFIS, SVM, LSSVM, ANN based MLP		MAPE, MPE, MBE, MSE, RMSE, nRMSE, R, R2	•				24 h
(Li et al., 2019)		•	•	MARS, ANN, LR, CRM	MARS	RMSE, RMSRE, MAE, MBE, R2	•		•		1 h
(Benali et al., 2019)		•	•	SP, ANN, RF	RF	RMSE, nRMSE, MAE, nMAE		•		•	1 h
(Walch et al., 2019a)	•	•		Bootstrapped ELM		MSE, R2	•			•	1 h/ Monthly/ Annual
(Fan et al., 2020b)	•	•	•	SVM-PSO, SVM-BAT, SVM-WOA, SVM, MARS, XGBoost	SVM-BAT	R2, RMSD, SI, MAE	•				24 h
(Feng et al., 2020)	•	•	•	PSO-ELM, ELM, SVM, GRNN, M5T, AE	PSO-ELM	MAE, RRMSE, NS	•			•	24 h/ Monthly/ Annual
(Fan et al., 2020a)		•	•	SVM, M5Tree, RF, XGBoost, CatBoost	SVM	R2, RMSE, NRMSE, MAE	•				24 h
(Marzouq et al., 2020)	•	•	•	HAEANN, SP, RT, RF	HAEANN	NRMSE; NRMSE		•		•	1 h
(Pang et al., 2020)		•	•	ANN, RNN	RNN	R2, RMSE, CVRMSE, NMBE,	•		•		10min/ 30min/1 h
(Lai et al., 2021)	•	•	•	DTC with FADF, SP, MLP, LSTM	DTC with FADF	RMSE, Rrmse, MAE, R2		•	•	•	1 h
(Zhou et al., 2021)	•	•	•	MTGPR, GPR	MTGPR	R2, RMSE, rRMSE, MBE,	•				24 h/ Monthly
(Puah et al., 2021)	•	•	•	RE-SOINN, PM, ES, ANN	RE-SOINN	MASE, RMSE		•			1min
(Bellido-Jiménez et al., 2021)		•	•	MLP, ELM, GRNN, SVM, RF, XGBoost	MLP, SVM, RF, XGBoost	MBE, RMSE, RRMSE, NSE, R2	•			•	24 h
(Narvaez et al., 2021)		•	•	LR, NN, RF, LSTM, GRU		MSE, R2, RMSE	•	•			1 h/24 h/ Weekly
(Husain and Khan, 2021)		•	•	SVM, KNN, RF, MLP, GPR	KNN, RF, SVM	MBE, RMSE, MPE, R, U95	•				24 h/ Monthly
(Guermoui et al., 2022)	•	•	•	GPR, GAO-LS-SVM, ELM, CNN-1D, CNN-GPR		R, NMAE, RMSE, nRMSE, MABE		•			1 h
(H. Wen et al., 2022)	•	•	•	FL-STANN, Lasso, ElasticNet, GRU, LSTM, attention-	FL-STANN	MAE, MBE, RMSE		•	•		5min

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Table 8 (*continued*)

Ref.	Information of prediction model					Evaluation indicators	Forecasting horizon				
	Hybrid model	Machine learning model	Compa-ration	Prediction model	Best model		Generalized	Multi-step ahead	Weather factor	Seasonal factor	Prediction horizon
				based LSTM, Transformer.							
(Ghimire et al., 2022b)	•	•	•	CXGBRFR, DNN, DBN, ANN, ELM, MARS	CXGBRFR	R, RMSE, MAE, WI, NSE, LM, Evar, SS, RMSESS	•			•	24 h/ Seasonal
(Haider et al., 2022)		•	•	Prophet, SARIMAX, ANN, CNN, LSTM		R2 score, RMSE, MAE, MSE	•	•		•	1 h
(Jia et al., 2022)		•	•	SVM, RF, GLMNET	SVM	nRMSE, nMAE, NMBE, R	•		•		24 h
(Ghimire et al., 2022a)	•	•	•	CSVR (1D CNN with SVR), LSTM, DBN, RBF, BRFR, MARS, WKNNR, GPML, M5T	CSVR	R, RMSE, MAE, RRMSE, RMAE, WI, NSE, LM, Evar	•			•	24 h
(Oyewola et al., 2022)		•	•	ANN with LM and SCG algorithm		MSE, R	•			•	24 h
(Nourani et al., 2022)	•	•	•	LSTM, sequenced-LSTM, wavelet-LSTM, FFNN		RMSE, DC, R2		•	•	•	1 h/24 h/ Monthly
(Geetha et al., 2022)		•	•	ANN with LM, RP and SCG algorithm	ANN with LM	MSE, R, MAD, RMSE, MAPE	•				1 h
(Goliatt and Yaseen, 2023)	•	•	•	CMAES, MARS, XGBoost		RMSE, MAE, NSE, R, VAF	•				24 h

In addition, a hybrid model integrated with several ML algorithms has been proposed to improve the prediction efficiency. Wang et al. (2017) proposed a hybrid method that combines WT and DCNN for forecasting PV power. The WT was used to decompose the original signal into several frequency series, and a DCNN was employed to extract the nonlinear features and invariant structures at each frequency. The hybrid model achieved a RMSE ranging from 1.356 to 6.336 for 45-min-ahead prediction. Li et al. (2020) proposed a hybrid WPD-LSTM network to forecast 1-h-ahead PV power. WPD was used to decompose the original series into subseries, and LSTM was developed for subseries to obtain the final forecasting results using a linear weighting method. The average RMSE and MAPE of the proposed model were 0.236 and 2.4, respectively. L. Wang et al. (L. Wang et al., 2022c) proposed a minute-hour-day-interval PV power prediction approach based on the hybrid LSTM-SVR-BO algorithm. They used EEMD to decompose the PV power time-series data, and then LSTM with the hyperparameters optimized by the BO method was integrated with SVR to predict the sub-series with different time intervals. The proposed model performed good accuracy with an average RMSE of 4.17%.

4.3. Large-scale rooftop solar photovoltaic potential forecasting

Studies of large-scale rooftop solar potential forecasting (Table 11, Table 12) were conducted based on the segmentation of building rooftop boundaries, to assess the application potential of PV panels on rooftops over large areas in cities. Typically, such studies focus not only on the meteorological parameters and solar radiation values, but also on the morphologies and geometries of cities and buildings. Table 11 (i.e., inputs and outputs, data resources, location and period) and Table 12 (i.e., prediction models, tools, evaluation indicators, forecasting horizons) collect the information of papers in the large-scale rooftop solar photovoltaic potential forecasting research.

CNN-based models are generally utilized to obtain rooftop segmentation from images, and evaluate the solar potential. Castello et al. (2019) and Huang et al. (2019) applied a supervised method called U-Net architecture based on CNN, to delineate rooftop solar panels and detect their sizes using pixel-wise image segmentation. The U-Net can automatically detect the area of a set of solar panels at the pixel level in test images with an accuracy of approximately 0.94. Zhong et al. (2021) proposed a deep learning-based method (DeepLab v3) to automatically extract the rooftop area with high-resolution satellite image segmentation with an accuracy of 0.92. This method enables the estimation of rooftop solar PV potential on a city scale. Sun et al. (2022) proposed a revised U-Net to estimate the spatial distribution of the rural rooftop photovoltaic power potential from publicly available satellite images with an overall accuracy over 92%. They also presented optimal tilt installation (OTI) and parallel installation (PI) approaches for calculating the installation potential of rooftop PV panels. Jiang et al. (2022) proposed a CNN and MLP to estimate rooftop PV power generation. They recognized the building footprints using satellite images and an automatic segmentation model, enabling the evaluation of the solar energy resource potential.

Many studies have combined machine learning models with other tools (GIS and LiDAR) to predict rooftop solar potential and explore the relationship between urban/building parameters and solar potential. Assouline et al. (2017) used SVMs and GIS techniques to estimate the rooftop solar PV potential for urban areas, including the estimation of monthly/annual solar radiation and PV electricity production potential on roofs. Mohajeri et al. (2018) used SVM classification in conjunction with GIS and LiDAR tools to classify 10,085 building roofs based on their received solar energy potential and areas suitable for PV installations in Switzerland. They can then provide design suggestions for optimizing solar panels on roofs and streamlining installation costs. A. Walch et al. (2019b) used LR, KNN, SVM, ELM, and RF models to estimate the annual solar irradiation on building roofs. Based on the comparison, RF had the best performance with an accuracy of 92%. They (Walch et al., 2020) further developed a methodology that combined RF, GIS, and physical models to estimate the PV potential for individual roof surfaces. They used the model to extract information from data and estimate the uncertainties with an MSE of 0.06–0.26, which were combined to obtain

Table 9
Index of papers for input value, output value and data information in spatial-temporal solar power forecasting research.

Ref.	Input value				Target value	Data resource			Location and period
	Meteorological parameter	Solar radiation	Historical solar power	Others		Collected	Simulated	Calculated	
(Hiyama and Kitabayashi, 1997)	T, WV, Time	Solar irradiation			Maximum PV generation	•			2 days in Japan
(Almonacid et al., 2014)	T (t, t+1)	Solar global irradiance (t, t+1) (1 h ahead)			PV power	•			Jaén (2011–2012)
(Wolff et al., 2016)	Time, T	CMV-based irradiance forecast (t), NWP-based irradiance forecast(t)	PV power (t-n, …, t) (15min-5h ahead)		PV power	•		•	921 PV systems in Germany (2012–2013)
(Wang et al., 2017)			PV power (t-n, …, t) (15min-2h ahead)		PV power	•			Elia (2015)
(Cervone et al., 2017)	CC, T, Time	Global horizontal irradiance		AZ, EL	PV power	•			3 power plants in Italy (2010–2013)
(Kumar and Kalavathi, 2018)	T, P, WS, AP, SH, RH,	Global horizontal irradiance, global diffused irradiance,			PV power	•			India (1 year)
(Wang et al., 2019)	WS, T, RH, WD	Global horizontal radiation, diffuse horizontal radiation (0.5–4years ahead)			PV power (5-day)	•			PV system in Alice Springs (2014–2017)
(Gao et al., 2019)	Ideal weather condition: T, RH, Tmax, Tmin (next day)	Ideal weather condition: daily solar irradiance (next day)	Non-ideal weather condition: historical PV power (2 day ahead)	Weather type	PV power	•			Solar plant in China (2016–2017)
(F. Wang et al., 2020a)			PV power (previous 3 days)		PV power (target day)	•			Desert Rock station in Nevada (2011–2016)
(Li et al., 2020)	T, WS, RH	Global and diffuse Horizontal Radiation	PV power (t-n, …, t) (1 h-ahead)		PV power	•			PV system in Australia (2014–2016)
(Zang et al., 2020)	NWP	Global and diffuse horizontal radiation (last 14 days)	Historical PV power (last 14 days)		PV power	•			Australia (2008–2018)
(Theocharides et al., 2020)	T, RH, WD, WS	Global irradiance (day-ahead)		AZ, EL	PV power	•			PV system in Cyprus (1 year)
(Kong et al., 2020)			PV power (1, …, t) (every 30s)	5 previous whole sky image (1, …,t) (every 30s)	PV power (t+1, …, t + n) (next 20min with 4min-step)	•			Sky Camara and PV system (2018–2019)
(Rana and Rahman, 2020)			Solar power (Pt-w, …, Pt) (5min-3h ahead)		Solar power (Pt+1, …,Pt + n) (n = 36 steps)	•			PV system in Australia (2015–2016)
(F. Wang et al., 2022a)	Time		Solar power of neighboring plant (t-n)	AZ, EL, Cim, CQ	PV power (t+15min to 4 h)	•	•		Satellite image and PV systems in China (2018–2019)
(Shams et al., 2021)				8 features correlated to wind and solar power units	Wind and solar power curtailments (WSPC)	•			PV plants in CAISO (2015–2020)
(Guermoui et al., 2021)			previous PV power (t-1, …, t-		PV power	•			3 PV power plants in

(continued on next page)

Table 9 (*continued*)

Ref.	Input value				Target value	Data resource			Location and period
	Meteorological parameter	Solar radiation	Historical solar power	Others		Collected	Simulated	Calculated	
			p) (15min, 30min, 45min, 60min ahead)						Algeria and Australia (1 year)
(Jebli et al., 2021)	T, WD, WS, RH, WS	solar radiation			Solar energy	•			Morocco (2016–2018)
(Chang et al., 2021)	T, Time	solar radiation			PV power	•			PV farm
(L. Wang et al., 2022c)	Time		PV power: 1-day ahead (5min), 7-days ahead (5min, 30min, 60min), 14-days ahead (60min)		PV power	•			2 PV plants in Arizona (2006)
(Etxegarai et al., 2022)	Time	Solar irradiation (previous 24 h with 10min-step)			Solar irradiation and generation	•		•	Euskalmet (2015–2017)
(Agga et al., 2022)	T, WS		previous PV power (48 h, 72 h ahead)	Power consumption	PV power (next 48 h, 72 h)	•			PV system in Rabat (2019–2020)
(X. Wang et al., 2022f)	T, WS, RH	Solar irradiance (day-ahead)			PV power	•			PV power station in China (2019–2020)
(J. Wang et al., 2022b)		Solar radiation (1-step, 2-step, 3-step ahead)			PV power	•			Belgium (2021)
(Zhang et al., 2022)	T, WS	Solar radiation previous twelve months (t-1, …, t-24)			Solar power, wind power (t)	•		•	Two large cascade hydropower plants in China (1981–2018)
(Rai et al., 2023)		Solar radiation			Solar power	•			Australia and India

the final PV potential.

In addition, some studies have directly utilized spatial data from GIS, LiDAR, and aerial orthoimagery to estimate the solar potential of city rooftops. Kouhestani et al. (Mansouri Kouhestani et al., 2019) and Pinna et al. (Pinna and Massidda, 2020) used a GIS and LiDAR tools to estimate the rooftop photovoltaic potential of buildings. The rooftop slope, azimuth, and shading were analyzed to select a suitable roof surface and evaluate solar radiation. Rooftop PV production and economic potential were also calculated. Martín-Jiménez et al. (2020) and Tiwari et al. (2020) combined LiDAR and aerial orthoimage data processing to geometrically characterize slope-level roofs and calculate their PV solar potential. They solved the extraction of roof boundaries and the automatic parameterization of roofs at the city, neighborhood, and building levels for different roof types.

5. Micro-scale solar energy forecasting

Micro-scale solar energy forecasting research focuses on solar potential forecasting for the existing urban environment and the parametric model (Table 13, Table 14), which both consider the interaction effect of morphological and geometric parameters of blocks and buildings on the solar energy performance, as shown in Fig. 5. Table 13 (i.e., inputs and outputs, data resources, location and period) and Table 14 (i.e., prediction models, simulation tools, evaluation indicators, forecasting horizons) collect the information of papers in the micro-scale solar energy forecasting research.

5.1. Solar potential forecasting in the existing urban environment

For the existing urban environment (Table 13, Table 14), solar potential forecasting was based on the parameters collected from existing blocks and buildings.

Studies have explored the relationship between block parameters and solar potential only using simulation tools for multiple instances of different types. Zhou et al. (2022) developed an estimation workflow for urban solar PV potential based on LiDAR data and building footprint. They delivered multiple results, including accurate geo-referenced 3D building models, annual solar irradiation maps, and classified roof segments. Tian (Tian and Xu, 2021) and Xu et al. (2021) proposed an evaluation approach for block solar potential and studied the effect of quantified morphological parameters on the solar potential of blocks in China. They discussed the distribution characteristics of solar potential in different types of blocks. They evaluated various strategies for the positioning of PV modules on block roofs and façades. Veisi et al. (Veisi and Shakibamanesh, 2022) provided an optimal model to evaluate the annual solar radiation at the city and block scales. Morphological variables at the city scale were studied, and solar radiation at different scales was investigated to verify the patterns of urban blocks for optimizing solar energy.

In recent years, ML and DL algorithms have been utilized to predict block solar potential. Lan et al. (2021) proposed a framework that utilizes urban streetscapes and the FCN model to estimate the solar potential, with a maximum error of less than 8%. Hosney Lila et al. (Hosney Lila et al., 2022) introduced an ANN model to predict the solar radiation for newly generated neighborhoods. The ANN was trained using a classified database of urban geometries, achieving a correlation of up to 0.99. Ren et al. (2022) proposed a novel 3D-GIS and DCNN. A 3D-GIS-based solar irradiance analyzer was developed to predict the dynamic rooftop solar irradiance by considering the shading effects of the surrounding buildings, and a DCNN was used to identify the rooftop availability. Lan et al. (2022) demonstrated a complete workflow using RF, naive Bayes, KNN, LDA, and ANN to understand the relationships

Table 10
Index of papers for prediction models, evaluation indicators and forecasting horizon in spatial-temporal solar power forecasting research.

Ref.	Information of prediction model					Evaluation indicators	Forecasting horizon				
	Hybrid model	Machine learning model	Compa-ration	Prediction model	Best model		Genera-lized	Multi-step ahead	Weather factor	Seasonal factor	Prediction horizon
(Hiyama and Kitabayashi, 1997)		•	•	ANN		MAE	•			•	1 h/24 h/ monthly
(Almonacid et al., 2014)		•	•	NAR, ANN		RMSE, R2		•		•	1 h
(Wolff et al., 2016)		•	•	SVR		RMSE, BIAS		•	•		15min-5h
(Wang et al., 2017)	•	•	•	WT-DCNN, BPNN, SVM, WT-SVM.	WT-DCNN	MAPE, RMSE, MAE		•		•	15min-2h
(Cervone et al., 2017)	•	•	•	AnEn + ANN, AnEn, ANN	AnEn + ANN	RMSE, CORR, BIAS	•		•		1 h
(Kumar and Kalavathi, 2018)		•	•	ANN, ANFIS	ANN	NRMSE	•				1 h
(Wang et al., 2019)	•	•	•	1D-CNN, LSTM, CNN and LSTM	CNN and LSTM	MAE, MAPE, RMSE, Loss		•			5min
(Gao et al., 2019)		•	•	LSTM with DGM, BP, LSSVM, WNN	LSTM with DGM	RMSE, MAD		•	•	•	15min/1 h
(F. Wang et al., 2020a)	•	•	•	LSTM-RNN, BP, SVM, persistent	LSTM-RNN	RMSE, MAE, COR		•	•		15min
(Li et al., 2020)	•	•	•	WPD-LSTM, LSTM, RNN, GRU, MLP	WPD-LSTM	MBE, MAPE, RMSE,		•	•	•	5min
(Zang et al., 2020)	•	•	•	ResNet, DenseNet, SVR, RFR, MLP, CNN.	ResNet, DenseNet	MSE, MAE, MSLE, MASE, NIA, U1, U2, MHE		•	•	•	1 h
(Theocharides et al., 2020)		•	•	ANN with linear regression		MAPE, nRMSE, SS		•			24 h
(Kong et al., 2020)		•	•	Cnn-LSTM, PredNet, convolutional LSTM		RMSE, nRMSE, MAE, MAPE		•	•		4min
(Rana and Rahman, 2020)		•	•	NN, SVR, RF, MLR	RF	MAE, MRE, S		•			5min
(F. Wang et al., 2022a)	•	•	•	Spatio-temporal correlation and neighboring information, SVM, GBDT, ARMA		RMSE, MAE		•	•		15min/1 h/ 2 h/4 h
(Shams et al., 2021)		•	•	RT, GBDT, RF, ANN, LSTM, SVR	RF	RMSE	•				1 h
(Guermoui et al., 2021)	•	•	•	IF-ELM		RMSE, nRMSE, MABE, nMAE, R		•			15min/ 30min/ 45min/ 60min
(Jebli et al., 2021)		•	•	LR, RF, SVR, ANN (MLP)	ANN, RF	RMSE, MAE, MSE, Max Error, R-squared	•				30min/1 h/ 24 h
(Chang et al., 2021)		•	•	TESDL, CNN, LSTM, MSTF	TESDL	Accuracy	•		•		1 h
(L. Wang et al., 2022c)	•	•	•	LSTM-SVR-BO, SVR, KNN, BPNN, GRU, Stacked-LSTM, LSTM, LSTM-SVR	LSTM-SVR-BO	RMSE, MAE, AbsDEV		•		•	5min/ 30min/ 60min
(Etxegarai et al., 2022)		•	•	LSTM, ANN, CNN	CNN	RMSE, MAE,		•	•	•	10min
(Agga et al., 2022)	•	•	•	CNN-LSTM, LR, KNN, DTR, CNN, LSTM, MLP	CNN-LSTM	RMSE, MAE, MAPE		•			24 h
(X. Wang et al., 2022f)		•	•	ETC, BRR, LR, GPR, MLP, SVR, GBR, RF, LAR, RR	ETC-RF, ETC-LAR, ETC-GBR, ETC-SVR	MRE, RMSE, MAE, MAPE, RMSE		•	•	•	15min
(J. Wang et al., 2022b)	•	•	•	GAvmd-Mogoa, Pso-LSsvm, Elm, LStm, CNN	GAvmd-MOgoa	RMSPE, MAE, RMSE, R2,		•		•	1 h
(Zhang et al., 2022)		•	•	MLP, SVR, RNN, EDT, Scenario-	Scenario-based	NRMSE, MAPE		•		•	monthly

(*continued on next page*)

Table 10 (*continued*)

Ref.	Information of prediction model					Evaluation indicators	Forecasting horizon				
	Hybrid model	Machine learning model	Compa-ration	Prediction model	Best model		Genera-lized	Multi-step ahead	Weather factor	Seasonal factor	Prediction horizon
(Rai et al., 2023)	•	•	•	based stochastic MPC CNN-BiLSTM, RNN, LSTM, GRU, CNN-LSTM	stochastic MPC CNN-BiLSTM	MSE, MAE	•				5 min

Table 11
Index of papers for input value, output value and data information in large-scale rooftop solar photovoltaic potential forecasting research.

Ref.	Input value				Target value	Data resource			Location and period
	Meteorological parameter	Solar radiation	Satellite image	Building parameter and others		Collected	Simulated	Calculated	
(Mainzer et al., 2017)	T	Irradiance profiles	Aerial images	BC	Building footprint, PV generation potential	•	•	•	49,573 buildings in Freiburg
(Assouline et al., 2017)	Lat, Long, H, SH, T, CC, P			Ag, At, BD, PD, Bt, Sf, Sn	weather maps, solar radiation, PV electricity production	•	•	•	1901 communes in Switzerland
(Mohajeri et al., 2018)	CC, T	Solar radiation		35 building features	Roof-shape classification	•	•	•	10,085 buildings in Switzerland, Meteorological data (1980–2000)
(Mansouri Kouhestani et al., 2019)	Lat, Long, CC, Trans	Solar radiation		Roof, BC, Sf	Rooftop PV electricity production and economic potential	•	•	•	55,877 buildings in Lethbridge, Meteorological data (2010–2014)
(Castello et al., 2019)			Aerial photos		Roof segmentation	•			700 images in Switzerland (2013)
(Huang et al., 2019)			Satellite images		Roof segmentation	•	•	•	180 high resolution images around the world
(Pinna and Massidda, 2020)	T, WS	Solar radiation		Ori, Roof, Sf	Global Horizontal Irradiation over the building rooftops	•	•	•	Meteorological Data (2007–2016)
(Martín-Jiménez et al., 2020)			Aerial orthoimagery	Aerial LiDAR data, BC, Ori, Roof	External roof edges and total roof area	•	•	•	Spanish
(Tiwari et al., 2020)			Orthophoto image	Digital surface model (DSM), Digital terrain model (DTM)	Classification of slope and aspect for building roofs, Rooftop solar radiation	•	•	•	Israel
(Walch et al., 2019b)	V	Horizontal irradiation		Roof, Ori, Sf, horizon maps	Rooftop solar irradiation	•			1,100,000 samples in Switzerland
(Walch et al., 2020)	Tmax, Rf, H	Global and direct solar horizontal radiation		Roof, Ori, Ag	Rooftop photovoltaic potential	•		•	38,000 roofs in Switzerland, Meteorological data (2004–2015)
(Zhong et al., 2021)			GES images		Rooftop segmentation, rooftop solar PV potential	•	•	•	20,000 samples in Nanjing, China
(Sun et al., 2022)			Satellite images	Ori	Rooftop segmentation, rooftop PV power potential	•		•	500 satellite images of 7 villages in China
(Jiang et al., 2022)	T, Time	Solar radiation	Satellite images	AZ, Roof	Rooftop segmentation, rooftop PV power potential	•	•	•	12,000 samples in China

between urban morphology and solar potential at an urban neighborhood scale using real 3D urban model data. The ANN model had the highest accuracy (0.86), and six urban morphology indicators, such as building density and plot ratio, were identified as influential in solar potential prediction.

Similar approaches have been applied to other forecasting research fields, such as energy demand (Silva et al., 2017), outdoor temperature (Yu et al., 2020), electricity consumption (Rosenfelder et al., 2021), and air pollutants (Huang et al., 2022). These results can provide inspiration for future research on micro-scale solar prediction.

Table 12
Index of papers for prediction models, evaluation indicators and forecasting horizon in large-scale rooftop solar photovoltaic potential forecasting research.

Ref.	Information of prediction model					Evaluation indicators	Forecasting horizon				
	Hybrid model	Machine learning model	Compa-ration	Prediction model	Tools		Genera-lized	Multi-step ahead	Weather factor	Seasonal factor	Prediction horizon
(Mainzer et al., 2017)		•		CNN	OverpassTurbo API	Accuracy	•				15min
(Assouline et al., 2017)		•		SVM	GIS, LiDAR	RMSE, RRMSE	•			•	Monthly/ Annual
(Mohajeri et al., 2018)		•		SVM	GIS, LiDAR	Accuracy	•				Annual
(Mansouri Kouhestani et al., 2019)					GIS, LiDAR	MBE, MAPE	•			•	Monthly/ Annual
(Castello et al., 2019)		•		U-Net architecture (CNN)		Accuracy	•				N/A
(Huang et al., 2019)		•		U-Net architecture (CNN)	Ladybug and Honeybee plugins in Grasshopper	MAE	•				Annual
(Pinna and Massidda, 2020)					GIS, LiDAR	MAE, RMSE	•			•	Monthly/ Annual
(Martín-Jiménez et al., 2020)					LiDAR, Aerial orthoimagery	Average error	•				N/A
(Tiwari et al., 2020)					GIS, LiDAR, Aerial orthoimagery	Accuracy	•			•	1 h/daily
(Walch et al., 2019b)		•	•	LR, KNN, SVM, ELM, RF		MAE, RMSE, MBE. R2	•				Annual
(Walch et al., 2020)		•	•	RF and ELM	GIS	RMSE, MAE, MBE, R2	•			•	1 h/ Monthly/ Annual
(Zhong et al., 2021)		•		DeepLab v3	GIS	Accuracy, Precision, Recall, F1 score	•			•	1 h/ Monthly/ Annual
(Sun et al., 2022)		•		The revised U-Net architecture (CNN)		Accuracy, IoU, F1-score.	•			•	Monthly/ Annual
(Jiang et al., 2022)		•		CNN and MLP	Global Solar Energy Estimator (GSEE)	R, RMSE	•			•	Monthly/ Annual

5.2. Solar potential forecasting with the parametric model and optimization

For the parametric model and optimization (Table 13, Table 14), the block and building parameters are usually set within a certain range to automatically generate parametric simulations, which enable prediction of solar potential and determination of optimal block parameters in cities. Vartholomaios (2019) proposed an ML approach to predict the annual, winter, and summer solar irradiation of urban surfaces. The author prepared the training dataset with parametric ray tracing simulations and predicted real blocks in Athens to capture the nonlinear relationships between solar irradiation and the predicted features. The RMSE of the prediction model ranged from 7.4 to 20.6 across different seasons. Shakibamanesh and Veisi (2022) performed a parametric simulation technique for facilitating planning in urban blocks using the Ladybug plugin in Grasshopper and calculated the solar radiation for different types of urban blocks. In addition, the maximum amount of solar energy was investigated based on the variables of the urban block types using Genetic Algorithms. They (Veisi et al., 2022b) determined the maximum annual solar radiation with the least volume in various patterns of urban blocks, and converted solar radiation simulation to emulation for various types of urban blocks. The Multi-Objective Optimization and ANN model were presented to characterize the influence of different urban scales on solar access and reach the optimal dimensions of the block. This analysis resulted in an accuracy improvement from 89% to 98%.

The parametric process of solar utilization potential forecasting for blocks and buildings can be achieved using Python to integrate the different workflows and fields. Chen et al. (Chen and Norford, 2017) developed an open Python library, Pyliburo, to provide or enhance the optimization capability for all design workflows. Makki et al. (2019) addressed the issue of generating urban morphological variations in response to conflicting criteria using a multi-objective evolutionary approach.

6. Building-scale solar energy forecasting

Fig. 6 shows the building-scale solar energy forecasting, which focuses on an individual building under the influence of the surrounding buildings and urban environment (Table 15, Table 16). Such studies focus on the influence of morphologies and geometries of buildings on the solar performance. Table 15 (i.e., inputs and outputs, data resources, location and period) and Table 16 (i.e., prediction models, simulation tools, evaluation indicators, forecasting horizons) show the information of papers in the building-scale solar energy forecasting research. Generally, a parametric approach is adopted to provide multiple scenarios and optimal suggestions.

Natanian et al. (2019) presented an automated parametric workflow for evaluating the impact of building parameters on PV potential and energy demand. J. Zhang et al. (2019) investigated the relationship between the urban block typology, solar energy harvesting potential, and building energy use efficiency for generic 3D building forms. They

Table 13
Index of papers for input value, output value and data information in micro-scale solar energy forecasting research.

Ref.	Input value				Target value	Data resource			Location and period
	Environmental parameter	Solar radiation	Block morphological parameter	Building parameter and others		Collected	Simulated	Calculated	
(Mavsar et al., 2019)	AZ	(Calculation)		Ag, Roof, Sf	Solar radiation, solar power	•		•	Slovenia
(Zhou et al., 2022)		(Simulation)		LiDAR data, building footprints	Solar irradiation map and PV power	•	•	•	Delft
(Lan et al., 2021)	SVF, SCF, Lat, Long	(Simulation)		fisheye image	Image segmentation, Solar yield	•	•	•	327 panoramic images in Queensland
(Hosney Lila et al., 2022)		(Simulation)	Bt	BH, Ag, Ori	Solar radiation	•	•		Aswan
(Tian and Xu, 2021)		(Simulation)	SF, AR, L/W, R/F, E/S, E/V, Slab, Tower	LA, BD, FAR, AH, BI	PV inatallation rate and potential, PV generation potential	•	•	•	36 residential blocks in Wuhan
(Xu et al., 2021)		(Simulation)	SF, AR,	L, W, AH, BD, FAR	Solar radiation potential, technical potential	•	•	•	48 cases of industrial blocks in Wuhan
(Ren et al., 2022)	3D-GIS data	(Simulation)		Satellite image	Available rooftop area, Solar energy potential	•	•	•	264 satellite images in Hong Kong
(Veisi and Shakibamanesh, 2022)	City scale: radar data, points cloud			Block scale: AH, pattern	Solar radiation	•	•	•	Kermanshah, Iran
(Lan et al., 2022)		(Simulation)	SF	BD, AH, Ori, Ag, At, FAR	Solar potential	•	•	•	534 urban mesh blocks in Adelaide
(Vartholomaios, 2019)	SVF	(Simulation)		Az, Ah	Solar radiation	•	•	•	30,000 parametric simulations in Athens
(Shakibamanesh and Veisi, 2022)		(Simulation)	V/A	AH, Ori, Dp	Solar radiation		•		Parametric simulations in Kermanshah, Iran
(Veisi et al., 2022b)		(Simulation)	Type	AH, Ori, At	Solar radiation, volume, area		•	•	Kermanshah, Iran

verified that urban form significantly influences the solar energy utilization and building energy consumption. Poon et al. (2020) analyzed the parametric relationship between 10 morphology parameters and the solar irradiance on rooftops and façades of buildings. The results showed that two sky exposure performance indicators, sky exposure factors (SEF) and sky view factors (SVF), have the strongest correlation with building irradiance for rooftops and façades. Chen et al. (2021) determined the vertical sky-diffuse solar irradiance on building facades based on 15 skies. Many simulations were conducted, and both ANN-based and regression-based models were employed to correlate irregular obstructions with the corresponding infinite-long cases of the same sky type. Krapf et al. (2021) presented a methodology for economic PV potential analysis using aerial images and CNN. The CNN was trained for segmentation of roofs and superstructures, achieving Intersection over Union values of 0.84 and 0.64. Through a parametric study, Wen et al. (J. Wen et al., 2022b) explored how the different designs of surrounding buildings affect the window-transmitted solar radiation energy (WTSRE) of the target building. Furthermore, they (J. Wen et al., 2022a) evaluated the effect of shading on the WTSRE performance for both individual target building and clusters in four typical climates in China. Design recommendations are provided to enhance the building energy efficiency in an urban context.

A few studies have focused on utilizing ML algorithms to assess the solar potential of buildings. Many studies have conducted parametric optimization analyses on daylight (Han et al., 2021; Kristiansen et al., 2022; Veisi et al., 2021; Wong et al., 2010), energy consumption and performance (D'Amico et al., 2019; Elbeltagi and Wefki, 2021; García Kerdan and Morillón Gálvez, 2022; Ilbeigi et al., 2020; Reynolds et al., 2018; Seyedzadeh et al., 2020) for buildings. These studies consider detailed environmental and building parameters as inputs into ML/DL models. Findings from these studies can inspire future research on building-scale solar predictions.

7. Discussion and outlook

7.1. Discussion and policy implications

Nowadays, all countries are vigorously developing the use of solar energy to achieve the goal of energy conservation and emission reduction. In order to effectively promote the application of solar radiation and photovoltaic power generation in urban environments, policymakers and stakeholders should focus efforts to promulgate relevant policies to respond.

For photovoltaic companies and projects, specific and favorable incentives from the state or government are very important. Economic and cost issues present significant barriers to the widespread adoption of solar photovoltaic technology in urban environments. Implementing certain technical and economic incentives can greatly promote the benign development of the photovoltaic industry. Likewise, consumers and users also require relevant policy measures, such as the photovoltaic grid-connected systems, storage issues, and photovoltaic electricity price subsidies. By improving relevant policies that prioritize the benefits for users, the practical application of photovoltaics in urban buildings can be effectively promoted. Furthermore, the research and development of photovoltaic technology calls for an exploration of various technical issues, such as enhancing the photoelectric conversion

Table 14
Index of papers for prediction models, evaluation indicators and forecasting horizon in micro-scale solar energy forecasting research.

Ref.	Information of prediction model					Evaluation indicators	Forecasting horizon				
	Hybrid model	Machine learning model	Compa-ration	Prediction model	Simulation tools		Generalized	Multi-step ahead	Weather factor	Seasonal factor	Prediction horizon
(Mavsar et al., 2019)					Polynomial	Accuracy	•			•	1 h/Annual
(Zhou et al., 2022)					LiDAR, GoogleMap	Accuracy	•				Annual
(Lan et al., 2021)		•		FCN	SunSPoT	Relative error	•				Annual
(Hosney Lila et al., 2022)		•		ANN	Ladybug plugins in Grasshopper	R2	•				Annual
(Tian and Xu, 2021)					Ladybug and Honeybee plugins in Grasshopper	R2, P value	•				Annual
(Xu et al., 2021)					Ladybug and Honeybee plugins in Grasshopper	R2	•				Annual
(Ren et al., 2022)	•			3D-GIS and DCNN (double U-Net)	3D-GIS, RayMan	R2, accuracy	•			•	1 h/Annual
(Veisi and Shakibamanesh, 2022)				Genetic Algorithms	Google Earth software, Ladybug plugin in Grasshopper	R2	•			•	Seasonal/ Annual
(Lan et al., 2022)		•	•	RF, naive Bayes, KNN, LDA, ANN	ArcGIS	TPR, FPR, accuracy	•				Annual
(Vartholomaios, 2019)		•		SVF, GBDT	Radiance, Ladybug plugins in Grasshopper	RMSE, MAPE	•			•	Seasonal/ Annual
(Shakibamanesh and Veisi, 2022)				Genetic Algorithms	Ladybug plugin in Grasshopper	R2	•			•	Seasonal/ Annual
(Veisi et al., 2022b)		•	•	ANN, Multi-Objective Optimization	Ladybug plugin in Grasshopper, Lunchbox	MSE, R2	•				Annual

rate and improving different materials. In terms of solar design, comprehensive assessment measures need to be conducted to guide solar radiation utilization and PV module installation in cities and buildings. The improvement of top-down policies and measures can enable policymakers and users to jointly promote the contribution of solar energy utilization to the whole society.

7.2. Limitations and future outlooks

This study reviewed important aspects of solar radiation and photovoltaic power forecasting in urban environments using ML and DL models at multiple scales, including the meso-scale, micro-scale, and building-scale. Overall, the literature on multiscale solar radiation and photovoltaic power forecasting research has different emphases and concerns. Although there have been many research reports demonstrating excellent performance in solar forecasting, there are still some unsolved issues.

(1) Meso-scale solar energy forecasting

For spatial-temporal solar radiation and solar power forecasting, the performance of the prediction model with ML/DL showed high accuracy and low error in the present review. All studies used appropriate prediction models and optimization algorithms to improve the forecasting performance for time-series datasets by combining different parameters and comparing diverse models. However, given the unstable characteristics of solar radiation, there still exist some issues that need to be addressed: 1) Solar prediction is complex because of the influence of the environment, weather, and PV operating conditions. Existing studies mainly predict solar radiation and PV power based on hourly data collected over several years, such as meteorological and geographical parameters. Therefore, future research should focus on incorporating detailed weather conditions, environmental parameters, sky factors, and air pollution factors into prediction models, as well as exploring different input combinations to optimize forecasting performance. 2) Existing literature mainly focuses on real-time hourly forecasting of solar radiation and PV power, which still faces unresolved economic issues. Hence, efficient time-series analysis for training and forecasting multiple tasks should be a key issue in the future. 3) Almost all studies analyzed the different performances of prediction models, which are generally considered black box models. The mathematical relationships between inputs and targets are not fully understood. Hence, a detailed or extended explanation of solar forecasting models is one of the main research directions.

For large-scale rooftop solar photovoltaic potential forecasting, CNN-based models are generally utilized to obtain rooftop segmentation from images, and then the solar potential is evaluated. The forecasting model, as well as technical tools (GIS, LiDAR, etc.), have been verified in existing studies for accurately recognizing rooftop boundaries, which are critical in assessing the solar potential of roofs. However, owing to the diversity of rooftops and the complexity of urban environments, there are still some issues that require further improvement: 1) The assessment of rooftop PV potential is directly influenced by the urban environment and construction conditions, such as roof obstacles and shading factors. Existing studies predict rooftop PV potential mainly based on the segmentation of roof boundaries from satellite images or other tools. Hence, considering mutual shading and accurately recognizing obstacles are necessary to obtain the available roof area for PV installation. These aspects are crucial for future advancements. 2) Urban buildings have diverse roof shapes and orientations, which directly determine the PV installation position. Therefore, it is important to determine the issue of large-scale rooftop PV potential forecasting by developing detailed PV installation approaches for flat or pitched roofs, as well as considering their orientation and tilt.

(2) Micro-scale solar energy forecasting

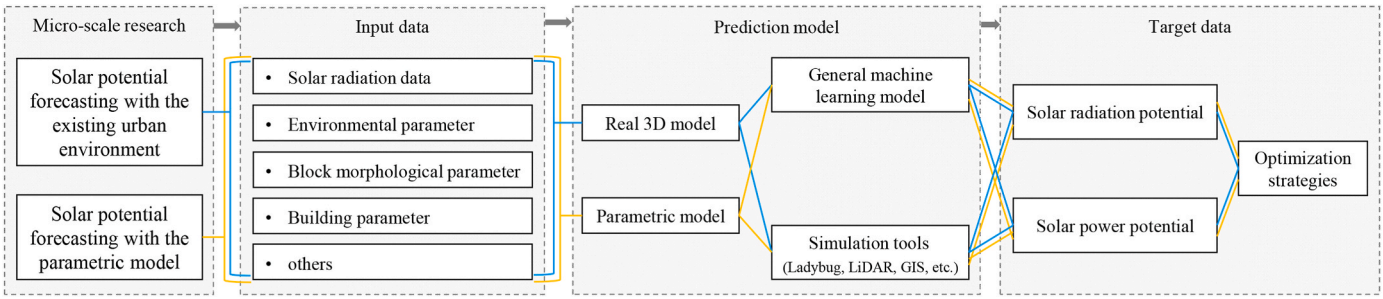


Fig. 5. Micro-scale solar energy forecasting research.

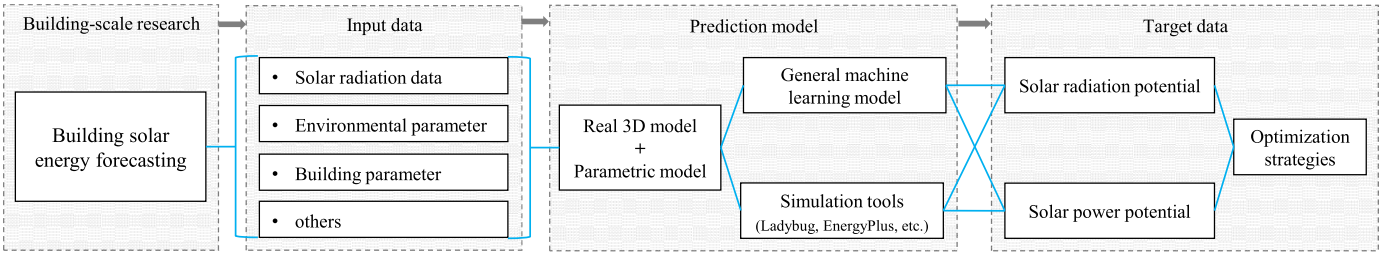


Fig. 6. Building-scale solar energy forecasting research.

Table 15
Index of papers for input value, output value and data information in building-scale solar energy forecasting research.

Ref.	Input value			Target value	Data resource			Location and period
	Environmental parameter	Solar radiation	Building parameter and others		Collected	Simulated	Calculated	
(Natanian et al., 2019)	Standard EPW file	(Simulation)	WWR, GP, BI, Ori, FAR, Type	PV potential, energy demand, daylight		•	•	1920 parametric iterations in Israel
(Zhang et al., 2019)		(Simulation)	9 parameters for solar potential (Type, BD, etc.), 9 parameters for building energy use (WWR, etc.)	PV potential, building energy use	•	•	•	Singapore
(Fuentes et al., 2020)		(Simulation)	UAV images, Ori	Solar radiation	•	•	•	Colombia
(Poon et al., 2020)	SVF, SEF	(Simulation)	10 parameters (AH, Ori, BD, etc.)	Solar irradiation		•	•	720 simulation cases in Singapore
(Chen et al., 2021)	EL	(Simulation)	5 parameters	15 standard skies, Solar irradiation		•	•	all locations world-wide (0.47 million parametric cases)
(Krapf et al., 2021)			Aerial images	Building outlines, PV power and economic potential	•		•	407 images in Germany
(Shi et al., 2021)			BD, Ag, block dimension	Solar energy potential and penetration	•	•	•	Singapore
(J. Wen et al., 2022b)		(Simulation)	Ori, BI, OA, L, AH	Window transmitted solar radiation energy		•	•	Beijing
(J. Wen et al., 2022a)		(Simulation)	Ori, BI, OA, L, AH	Window transmitted solar radiation energy		•	•	Harbin, Beijing, Shanghai, Guangzhou

For solar potential forecasting with the existing urban environment and parametric model, most studies simulated solar radiation of different block types with 3D-models. This helps establish the relationships between block parameters and solar utilization potential, and propose strategies for block and building design. It makes sense to combine ML/DL with solar potential forecasting for blocks and buildings because a well-developed prediction model can provide recommendations for solar utilization at the beginning of design. A few studies used ML/DL to explore solar potential forecasting models and obtained good prediction performance. However, few studies have adopted ML/DL due to various challenges, such as dataset collection, the mutual influence of buildings, the design of prediction models, and the diversification of prediction results. 1) Few studies have used the ANN model, while most research does not combine the analysis with ML/DL prediction models. This is because of the difficulty in obtaining dataset resources. This research mainly focuses on the analysis of urban buildings and the environment, which typically involves the use of simulation software to obtain solar radiation values. This means that the author needs to create

Table 16
Index of papers for prediction models, evaluation indicators and forecasting horizon in building-scale solar energy forecasting research.

Ref.	Information of prediction model					Evaluation indicators	Forecasting horizon				
	Hybrid model	Machine learning model	Compa-ration	Prediction model	Simulation tools		Generalized	Multi-step ahead	Weather factor	Seasonal factor	Prediction horizon
(Natanian et al., 2019)					Ladybug and Honeybee plugins in Grasshopper, EnergyPlus	LM	•			•	1 h/Daily/ Monthly
(Zhang et al., 2019)					Ladybug and Honeybee plugins in Grasshopper, EnergyPlus	R2, P value	•				Annual
(Fuentes et al., 2020)			•	UAV-DSM	Solar Radiation Toolset and GIS, PVSyst	Accuracy	•			•	1 h/ Monthly/ Annual
(Poon et al., 2020)					Ladybug plugin in Grasshopper	R2, P value, Accuracy	•				Annual
(Chen et al., 2021)		•	•	ANN	RADIANCE	R2, MBE, RRMSD	•		•	•	1 h
(Krapf et al., 2021)		•	•	CNN		IoU	•				Annual
(Shi et al., 2021)		•		K-means clustering method	GIS, Urban Block Generator in Grasshopper, City Energy Analyst	aCAPEX	•				1 h/Annual
(J. Wen et al., 2022b)					EnergyPlus, MOOSAS-FastSolar	Percent Error	•				1 h
(J. Wen et al., 2022a)					EnergyPlus	Shading Factor (SF), Impact Factor (IF)	•				Monthly

their own research dataset, which involves the simulation of 3D models for existing blocks or parametric models. This process is time and effort-intensive, thus efficient solutions are needed to address the challenge of abundant urban building models and streamline the simulation procedures. Additionally, integrating more ML/DL models with solar potential forecasting is vital for future research. 2) Urban blocks are constructed using parameters such as urban morphology, layout, orientation, type, height, and building density. Hence, research on the relationship between building parameters and solar potential forecasting is a key issue in the future. 3) Existing studies have mainly focused on solar potential forecasting for the entire blocks or buildings. Hence, quantifying the solar potential prediction for both the roof and four façades with the influence of surrounding environments and buildings, as well as determining the detailed PV installation position on the roofs and façades, need to be further quantified and prioritized in the future. 4) Existing studies predict the solar potential within annual and seasonal periods. Solar energy has time sequence characteristics. Thus, determining the detailed period analysis (hourly and daily) of the solar potential for blocks and buildings needs to be explored. 5) Different regions, locations, and building types exhibit diverse solar distributions and utilization patterns. It is worth comparing their differences and providing precise strategies for the micro-scale solar potential based on specific locations and building functions. 6) Economic factors, as well as energy conservation and emission reduction targets, are crucial factors in PV installation. Hence, future research also needs to predict the solar potential by considering factors such as the balance of payments, economic payback period, and carbon emission reduction.

(3) Building-scale solar energy forecasting

For building-scale solar energy forecasting, the existing issues and possible future research directions are similar to those mentioned in micro-scale solar energy forecasting because the research object changes from multiple buildings to single buildings in cities. In addition, there are some issues that need to be explored: 1) It is necessary to train a prediction model for the solar potential of a single building with the influence of surroundings. Subsequently, the model should undergo randomized testing in real building neighborhoods, facilitating a validation and correction process to evaluate its forecasting performance. 2) Solar potential forecasting and comparison for different functional types of buildings (i.e., residential, commercial, industrial buildings, etc.) need to be explored in the future, owing to their different settings for building parameters.

In addition, the future use and efficient development of solar energy requires joint efforts of many valuable fields, including the development of green hydrogen (Gado and Hassan, 2023; Garud et al., 2023; Herdem et al., 2023; Juárez-Casildo et al., 2022; Lee et al., 2022; Rioja-Cabanillas et al., 2023; Temiz and Dincer, 2022), grid-connected PV system (Argyrou et al., 2021; Elkadeem and Abido, 2023; Trillo-Montero et al., 2014), etc. Accurate solar forecasting can promote the diversification of solar energy development.

8. Conclusions

Machine learning and deep learning models are among the most popular research topics for solar prediction, owing to their high accuracy and wider application. This paper provides a comprehensive review of ML- and DL-based multiscale solar radiation and photovoltaic power forecasting research in urban environments. It covers various analysis approaches, prediction models, and the characteristics of datasets from the meso-scale, micro-scale, and building-scale viewpoints. This paper also discusses several existing issues and possible future research directions for multiscale solar prediction in cities using ML and DL algorithms.

Regarding solar radiation and photovoltaic power forecasting, the research analyzes multiple time-series data at different spatial scales, each with its specific concerns and research approach. Hence, a multi-scale methodology is proposed, classifying the evaluation of prediction effectiveness in diverse research scenarios into meso-scale, micro-scale, and building-scale. For meso-scale solar forecasting, spatial-temporal solar radiation forecasting research is mainly related to meteorological and geographical parameters, while solar power forecasting research is generally influenced by solar radiation and meteorological parameters. They have a strong relationship with each other, and both have the characteristics of a time sequence. Large-scale rooftop solar photovoltaic potential forecasting research mainly focuses on PV applications on rooftops, which are influenced by environmental and building parameters. For micro-scale and building-scale solar forecasting, these are two smaller scales that consider the interaction between buildings, such as building height, interval, shading and volume factors. Detailed solar potential predictions for blocks and buildings were conducted using existing urban models and parametric models, to provide optimization strategies for site planning and building design. However, these two smaller scales have received limited attention in solar potential forecasting research using ML and DL methods, whereas other fields such as energy consumption have conducted detailed investigations. Therefore, it is possible to explore the solar potential of urban buildings using ML approaches.

Although many studies have been conducted in the literature, further work is needed to fully explore the practical utilization and multiple development of solar radiation and photovoltaic power forecasting in diverse urban environments. This study demonstrates that each forecasting model and approach has its concerns and advantages in multi-scale research. Furthermore, these analyses can help researchers and engineers in identifying suitable ML/DL algorithms and prediction approaches, to improve their specific prediction tools. Consequently, this exploration emphasizes the potential of artificial intelligence in solar radiation and photovoltaic power forecasting across various urban scenarios. This can promote the utilization of solar energy in cities, as well as contribute to the urban low-carbon growth.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Nomenclature

Symbols	
ADC	Average Daily Cloudiness
ADD	Average Day Duration
Ag	Building Ground Floor Area
Ah	Horizontal Angles
AP	Air Pressure
AQI	Air Quality Index
AR	Aspect Ratio
At	Total Building Ground Floor Area
ATDR	Average TOA Daily Radiation
AZ	Solar Azimuth
BC	Building Coordinates
BD	Building Density
BH	Building Height
BI	Building Interval
Bt	Building Typology
CC	Cloud Cover
CI	Clearness Index
Cim	Cloud Images
CMV	Satellite-Based Cloud Motion Vectors
CQ	Cloud Quality
CR	Cumulated Rainfall
CSD	Clear Sky Days
Dco	Diffusion Coefficient
Df	Diffuse Fraction
Dp	Building Dimension Proportions
E/S	Envelope-to-Site Area Ratio
E/V	Envelope-Volume Ratio
EL	Solar Elevation
ER	Extraterrestrial Radiation
Et	Integral of Temperature Values
Evap	Evaporation
FAR	Floor Area Ratio
GIS	Geographic Information Systems
GP	Glazing Properties
H	Altitude
HDD	Heating Degrees Day
hfls	Surface Upward Latent Heat Flux
hfss	Surface Upward Sensible Heat Flux
Hmax	Maximum Weather Humidity
Hmin	Minimum Weather Humidity

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Symbols	
Hs	Near Surface Specific Humidity
Ics	The Theoretical Clear-Sky DNI
L	Building Length
L/W	Length-to-Width Ratio
LA	Land Area
Lat	Latitude
LiDAR	Light Detection and Ranging
LM	Load Match Index
Long	Longitude
N	Maximum Possible Sunshine Hours
NWP	Numerical Weather Predictions
O3	Ozone
OA	Offset Angle
Ori	Orientation of The Building Rooftops
P	Precipitation
P/Po	Surface Air Pressure Ratio
PD	Population Density
R/F	Roof-to-Façade Area Ratio
R/Ro	Relative Humidity Ratio
Rain	Rainfall
RD	Rainy Days
Rf	Surface Reflectance
RH	Relative Humidity
Roof	Rooftop Slope
S/So	Sunshine Duration Ratio
SCF	Sun Coverage Factor
SEF	Sky Exposure Factor
Sf	Fully Shaded Areas
SF	Shape Factor
SH	Sunshine Hours
Sn	Non-fully Shaded Areas
SR	Sunshine Rate: The ratio of sunshine hours to maximum sunshine duration
SVF	Sky View Factor
SZA	Solar Zenith Angle
T	Temperature
T/To	Temperature Ratio
TAI	Top of Atmosphere Insolation
Tave	Average Temperature
TD	Temperature Diurnal: Tmax-Tmin
Tdew	Dew-Point Temperature
Time	Period (Month/Day/Hour)
Tmax	Maximum Temperature
Tmin	Minimum Temperature
Trans	Transmittivity
Twet	Wet Bulb Air Temperature
V	Visibility
V/A	Volume to Area Ratio
VPD	Vapor Pressure Deficit
W	Building Width
wap	Omega (Lagrangian Tendency of Air Pressure)
WD	Wind Direction
WS	Wind Speed
WV	Wind Velocity
WWR	Window to Wall Ratio
Abbreviations	
AE	Autoencoder
AnEn	Analog Ensemble
ANFIS	Adaptive Neuro-Fuzzy Inference System
ANFIS-ACO	ANFIS Optimized with Ant Colony
ANFIS-GA	ANFIS Optimized with Genetic Algorithm
ANFIS-PSO	ANFIS Optimized with Particle Swarm Optimization Algorithm
ANN	Artificial Neural Network
AR	Typical Linear Autoregressive Model
ARMA	Auto-Regressive and Moving Average Model
Bat	Bat Algorithm
BO	Bayesian Optimization
BP	Back Propagation
BRR	Bayesian Ridge Regressor
CatBoost	Gradient Boosting with Categorical Features Support
CMAES	Covariance Matrix Adaptive Evolution Strategies
CNN	Convolutional Neural Networks
CRM	Cloud-Cover Radiation Model
CXGBRFR	CNN and Xgboost With RF
DBN	Deep Belief Network
DCNN	Deep Convolutional Neural Network
DGM	Discrete Grey Model

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Symbols	
DHI	Diffuse Horizontal Irradiation
DNI	Direct Normal Irradiance
DNN	Deep Neural Networks
DTC	Deep Time-Series Clustering
EDT	Ensemble Decision Tree
EEMD	Ensemble Empirical Mode Decomposition
ELM	Extreme Learning Machines
ES	Exponential Smoothing
ETC	Extremely Randomized Trees Classification
FADF	Feature Attention Based Deep Forecasting
FCN	Fully Convolutional Network
FFNN	Feed-Forward Neural Network
FL-STANN	Federated Learning Spatial–Temporal Attention-Based Neural Network
GAO-LS-SVM	Grasshopper Optimization Least Square Support Vector Machine
GAvmd	Variational Mode Decomposition Based on Genetic Algorithm
GBDT	Gradient Boosting Decision Trees
GBR	Gradient Boosting Regressor
GHI	Global Horizontal Irradiation
GLMNET	Generalized Linear Modeling
GMDH	Group Method of Data Handling Type Neural Network
GPR	Gaussian Process Regression
GRNN	Generalized Regression Neural Networks
GRU	Gated Recurrent Unit
GSR	Global Solar Radiation
HAEANN	History and Architecture Evolutionary Artificial Neural Networks
IF	Iterative Filtering Decomposition Method
KNN	K-Nearest Neighbors
LAR	Lasso Regressor
LDA	Linear Discriminant Analysis
LM	Levenberg-Marquardt Algorithm
LR	Logistic Regression
LSSVM	Least Square Support Vector Machine
LSTM	Long Short Term Memory Networks
M5T	M5 Model Tree
MARS	Multivariate Adaptive Regression Spline
MLFFNN	Multilayer Feed-forward Neural Network
MLP	Multi-layer Perceptron
Mogoa	Multi Objective Grasshopper Optimization Algorithm
MPC	Model Predictive Control
MSTF	Multiple Short-term Functions
MTGPR	Multi-task Gaussian Process Regression
NAR	Nonlinear Autoregressive Neural Network
PM	Persistence Model
PredNet	Predictive Coding Network
PSO	Particle Swarm Optimization Algorithm
RBFNN	Radial Basis Function Neural Network
RE-SOINN	Regression Enhanced Self-organizing Incremental Neural Network
RF	Random Forest
RNN	Recurrent Neural Network
RP	Resilient Back Propagation
RR	Ridge Regressor
RT	Regression Trees
SARIMAX	Seasonal Auto Regressive Integrated Moving Average
SCG	Scaled Conjugate Gradient Algorithm
SP	Smart Persistence
SVM	Support Vector Machine
SVR	Support Vector Regression
SVR-poly	Polynomial Basis Function
SVR-rbf	Radial Basis Function
TESDL	Traditional Encoder Single Deep Learning
UAV-DSM	Unmanned Aerial Vehicles–Digital Surface Model
WKNNR	Weighted K-Nearest Neighbor
WNN	Wavelet Neural Network
WOA	Whale Optimization Algorithm
WPD	Wavelet Packet Decomposition
XGBoost	Extreme Gradient Boost

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